

Unit 17BC:

Design & Develop a Mobile App

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Assignment Context & Brief Requirements:

“Educational apps are frequently changing and students are increasingly trying to find tools to help manage their work and study in college. You are going to create an app to support students with their study skills and independence. This app can be an idea partially of your choice as long as it meets the criteria below.

To ensure your app is something other students will want to use you will need to research your concept and plan it carefully before moving on to develop and test it.

Your app must:

- Help support students with their general study skills (not subject specific)
- Utilise gamification to help motivate them to engage with it
- Be visually appealing
- Intuitive to use and not require additional guidance
- Run on an Android or iOS mobile device”

These are requirements that I will need to meet with my app’s design.

Introduction

I am going to design create a pomodoro style timer app. The purpose of my app will be to support students with their study skills and independence; It will allow them to manage their time effectively, without needing any help from others.

A pomodoro timer splits up time into chunks of focus sessions and breaks. It helps people work more efficiently as they work with high concentration for short bursts of time (focus sessions, usually 25 minutes) and are then rewarded with breaks in between (usually 5 minutes). After they have completed 4 (typically) focus sessions, they are rewarded with a longer break (usually 45 minutes).

Students will be able to use my app to assist them with managing time around homework, coursework, and studying. Because of this, they will spend their time more effectively and free up time for other things, such as hobbies they might enjoy or spending time with friends & family. Furthermore, this will improve their overall mental health as it will help them balance their life.

My App's Design

Requirements

My app has several functional and non-functional requirements. Below are the requirements/features I want to include in my app.

Functional:

- **Focus Timer** – *A functional timer for focus sessions.*
- **Break Timer** – *Break/rest timer between focus sessions.*
- **Long Break Timer** – *Long break timer after 4 focus sessions*
- **Adjustable timers** – *The three above timers, and the ability to adjust the duration of them, is necessary. This is what users (students) will follow when studying, revising, or working towards college. Furthermore, by having the above timers, it would help my app meet the [brief requirement](#) of utilising gamification, as a countdown timer is a common element of games.*
- **Notifications, Alarms, & SFX** – *This will notify the user that the current timer has ended; This is an important feature as users might not notice the timer has ended, when they need to stay on track and time. SFX is another feature that would add to the gamification of my app, hence meeting the [brief requirement](#) further.*
- **Pause button** – *Allows the user to pause the current timer; This is important as the user may need to stop working for an important reason, such as going to the toilet, doing a more important/urgent task, or in the event of an emergency.*

Non-Functional:

- **Clear to read UI** – *Clear and large enough text will mean users can easily read any text on my app. This is especially important for those who may have difficulty reading or visual impairments. Alongside this, it will also assist my app to meet the [brief requirement](#) of my app being visually appealing.*
- **Clear and simplistic layout** – *By only including necessary features (including text and buttons), users will not get overwhelmed; Therefore, they will be able to focus better when using my app, helping them concentrate and improve their study skills. By meeting this requirement, my app will also meet the requirement of being visually appealing (as stated in the [assignment brief](#)).*

- **Intuitive to use** – *This ties in with having a clear and simplistic layout. It should be obvious to the user(s) on what things do, such as what each button does or text means. By being intuitive to use, it will improve the user's independence (toward studying) and remove the need for additional guidance. Furthermore, this requirement is also within the [assignment brief](#), making it an important requirement to meet.*

Extra Features:

Depending on how long it takes me to develop my app, I will try and implement these features:

- **Colour changing background** – *The app's colour theme could have shades of red during the focus sessions and change to shades of blue during the breaks in between the focus sessions. This gives a visual representation of what session the user is in, while also giving psychological responses to the colours (for more detail, see App Theme & Style, Colour Scheme). By having a background that changes colour based on what session the user is in, it will help meet the [brief requirement](#) of my app being visually appealing; Furthermore, it would make my app more intuitive to use (meeting another requirement) as it will be very clear to the user what session they are in, reducing confusion.*
- **Skip buttons** – *Allows users to skip to the end of the session they are in (focus or break). This gives the user more control over their studying, which is more appealing to the user. Furthermore, if they need to finish early, they can. On the other hand, users could abuse this to not use the app effectively for their studies. As skip buttons are used in a lot of mobile apps, such as music playing apps (that students will likely be familiar with), this will be an intuitive-to-use feature; Therefore, this feature will assist my app meet the [brief](#) (and [my own](#)) requirement of my app being intuitive to use.*

Gamification Features:

To meet the [brief requirement](#) of utilising gamification to help motivate the user to engage with my app, I will add some of these features:

- **Focus Streak** – *A streak that is maintained by the user completing focus sessions without skipping one (using the skip button mentioned in Extra Features); This would encourage users to complete full focus sessions, reducing the issue of having Skip*

buttons. Alternatively, a streak could be maintained by completing a focus session once a day; This feature would encourage users to come back to the app repeatedly.

- **User Praise** – *When the user completes focus sessions, they receive an on-screen message congratulating or praising the user. This will help motivate users to continue working hard, hence encouraging students to continue to study independently and efficiently. By giving students positive reinforcement, it will entice them to use my app again, as it gamifies my app; This feature would assist my app in meeting the [brief requirement](#) of utilising gamification to help motivate the users to engage with my app.*
- **Unlockable Customisation** – *Unlockable customisation for the background, timer and colours. This feature would assist my app in meeting the [brief requirement](#) of utilising gamification. However, this would be quite a hard feature to implement as I would likely have to create an in-game currency and a shop. Because of this, I am unlikely to be able to include this feature within the given time frame.*

App Theme & Style

Colour Scheme

The colour scheme for my app needs to be visually appealing (for my app to meet the [assignment brief](#) of being visually appealing). It also needs to have colours with connotations of calmness, focus and or concentration to keep the user motivated and engaged. I am going to use [Coolors](#) to choose a colour scheme for my app.

Chosen Colour Scheme: “Fiery Ocean”

For my app I am going to use the colour palette “Fiery Ocean”.

This colour palette is one of the most popular ones on the Coolors website and is currently “trending”. By using a popular and trending colour palette, the visual design of my app is likely to be appealing, given it has a pleasing layout as well. This will allow me to meet the [brief requirement](#) of my app being visually appealing.



I preferred this colour scheme over others as it had both shades of blue *and* red. This is perfect for my app as I can use the red for the focus sessions and the blue for the break sessions. Because of this, I can implement my extra feature idea of a “Colour changing background”, ensuring my app meets the gamification requirement stated in the [brief](#) (for more information see [here](#)).

Another reason I chose this colour scheme, was because it has some appealing shades of blue and red, alongside an off-white colour. As mentioned before, I will use different colours for my app during different sessions (focus and break), I have gone into more detail about this below:

Use in Focus Sessions:

I will use the shades of red for the focus sessions, as it has connotations of energy, urgency, and action, to help focus and motivate users (students) on their work (such as homework, revision, or coursework).

Use in Break Sessions:

For the break sessions I will use the shades of blue, as blue has connotations of calmness and tranquillity; This will help calm and relax users during their breaks

between focus sessions. The contrasting change between the two colours should help the user focus again when the colour scheme of the app changes back to the shades of red.

Use throughout the app:

I will use the off-white colour throughout the app, not specifically in the focus or break sessions. I will most likely use this for the colour of interactable or important UI, such as important text and buttons; This will make elements stand out to the user and be intuitive that they are interactable or important.

Use of Colour Connotations:

By using colours in my app that have specific connotations, the students who use my app could have psychological responses that will engage and focus them on their work and calm and cool them down during breaks. *I have explained my use of colours on the previous page.*

Simplistic Colour Scheme Purpose:

By only using a few colours at a time, it will reduce the chance of distracting users from their work (such as homework or revision for students). Furthermore, the colours are not overly dark or light, this ensures they do not stand out too little or too much.

Instead, they have a mixture of attractive lighter and darker shades; This will make my app visually appealing and will allow me to make certain elements of my app stand out, such as buttons, important text, and the focus/break timers.

Typography

I am going to create my app using [Thunkable](#). Because of this, I need to choose from fonts that are built into Thunkable.

I have selected three fonts that I would like to use for my app:

Font 1 – Oswald:

Oswald is very clear to read; Because of this, users will be able to read it very clearly. This is good as those who may have difficulty reading will be able to read it easier than other fonts. It also has thicker letters than other fonts, this would make it good for subtitles or important text.

A sample of the Oswald font, showing the words "Oswald Font" in a bold, sans-serif typeface. The text is contained within a light yellow rectangular box with a thin black border.

Furthermore, it looks simplistic without looking boring; This means it will not distract the user but will still be visually appealing (meeting the [requirement](#)).

Font 2 – Poppins Font:

Similarly to Oswald, Poppins is clear to read. It is more spaced out than Oswald and each letter is thinner, which makes it clearer to read.

A sample of the Poppins font, showing the words "Poppins Font" in a modern, sans-serif typeface with wide letter spacing. The text is contained within a light yellow rectangular box with a thin black border.

Again, it is an appealing font while remaining simplistic; Therefore, it will not distract the user.

Font 3 – Pacifico Font:

Pacifico is fancier and more stylish than the other two fonts, however, this is slightly harder to read. Because of this, when used, the font size should be larger to ensure readability.

A sample of the Pacifico font, showing the words "Pacifico Font" in a cursive, script typeface. The text is contained within a light yellow rectangular box with a thin black border.

Because of this, I am most likely to use this font as a title or as subtitles.

However, as this font may be too hard for some users to read, which would go against my requirement of my app having a [clear to read UI](#), I am unlikely to use it in my app.

Accessibility

I need to consider different accessibility needs, including Colour Blindness, Dyslexia and hearing problems.

- **Colour Blindness & Visual Impairments** – To make my app colour blind friendly, I will ensure that any text has a high contrasting background so it can clearly be read. This will also be beneficial for users who may have visual impairments. Furthermore, to ensure text is visible to users (including those with visual impairments), I will make sure the font sizes are large enough to be read with ease. Additionally, by having clear to read text on contrasting background, it will help users with dyslexia read the on-screen text. By ensuring this accessibility in my app, I will meet my [requirement](#) of it having a clear to read UI,
- **Navigation/Input** – I will need to ensure my app's navigation and user input is easy to understand and reduce complex gestures. This will ensure that my app meets the [brief](#) and [my own](#) requirements of being intuitive to use without needing additional guidance. To do this I will ensure that the app's navigation is clear and easy to use with text labelling what elements of my app do. An example of this would be labels on buttons (e.g.: a button opening a menu would be labelled “Menu” or a button leading to a previous screen labelled “Back”). Furthermore, I will ensure interactable elements are large enough to easily be used on a mobile device.
- **Orientation** – As a possible, additional feature, I could create the app in a way that supports a portrait and landscape layout. This gives the user more choice on how they want to use the app. Furthermore, for users on devices such as tablets, my app would look more appealing with a landscape layout, whereas someone on a phone would likely prefer a portrait layout. This would support my app's ability to run on an Android or iOS mobile device (meeting the [requirement](#)), however it is not a necessary feature to run on these mobile devices.

Layout

To design the layout of my app, I have used [Balsamiq Wireframes](#) to create wireframes for each screen of my app.

Focus Session screen:

I have created a focus session screen. As explained in App Theme & Style, I have used the shades of red alongside the off-white colour. On top of this, I have also used black to ensure important text and buttons stand out and are clear to read (to meet my [requirements](#) of having a clear to read UI, and the user not needing additional guidance).

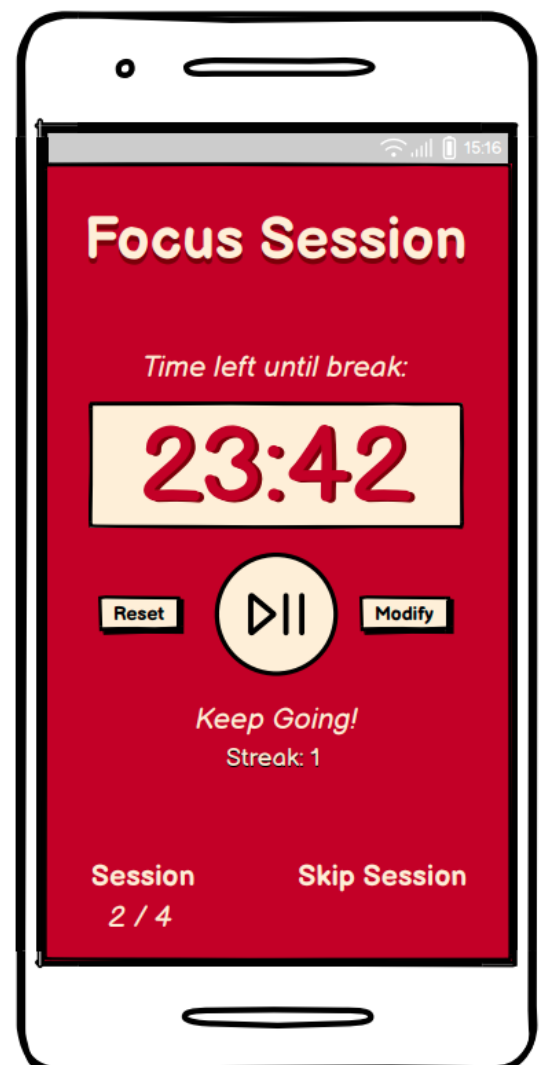
I have put text at the top to clearly state that the user is in a focus session, not a break. This ensures the user will not get confused over what session they are currently in (ensuring the user does [not require additional guidance](#)).

I have also put text and a timer relating to when their break will start “Time left until break”. The timer has been placed in the centre of the screen as it is one of the most important pieces of information to the user.

The user can press a “Reset” button to reset the timer and can press a “Modify” button to take them to a screen where they can modify timer durations (see *Modify Screen*). The reset button will pause the timer and reset it to the duration before it started.

There is also a play/pause button which allows the user to pause and resume the current timer (whether that be during a Focus or Break session). I have placed this below the timer itself, as it is directly related to it. Furthermore, play/pause buttons in apps for timers are commonly placed right below the timers. By using predictable placements of elements in my app, such as this, it will make my app more intuitive to use (hence meeting the [requirement](#)). This is needed as my app is supposed to support students with their independence (alongside study skills); By having an intuitive-to-use app, it will improve user’s independence as they will not need help to use it.

Focus Session screen



Below the play/pause button, I have put a short motivational message for the user “Keep Going!”. By having encouragement in my app, it will motivate users to continue working hard and, in this case, help motivate students to keep studying.

I have also placed a session counter in the bottom left to indicate to the user what session they are on. Once they complete the final session, they are rewarded with a long break (45 minutes by default, compared to the normal break which is 5 minutes by default); By having this reward for working effectively, it gamifies the management of work and entices the user to continue to study & use the app. This is very important for students as without this gamified aspect to my app, they may lose interest in their work and studies towards college.

I have also created a streak counter just below the mini motivational message. This will again encourage the user to work; By using gamification, in this case having a streak (*as described in [Requirements](#)*), it entices the user to continue working and reduces the chance of them losing interest in their work.

Finally, I have placed a “Skip Session” button in the bottom right corner. This will allow the user to skip the current session they are in (either Focus or Break). If used during the focus session, it will reset their current streak (I may implement a warning notification to alert the user of this as it may not be obvious).

Break Session screen

I have also created a break session screen. This screen is shown after a focus session is completed (or skipped).

Alike the Focus Session screen, I have used the shades of blue for the same reason identified in App Theme & Style; By having the app's colour scheme change between sessions, it makes it clear to the user that it is no longer a focus session and instead a break/rest. Furthermore, the colours chosen can have calming psychological effects on users (see [App Theme & Style for more information](#)).

On this screen I have also put elements in the same position as the Focus Session screen. These elements include:

- Session Title (at top middle)
- "Time left until focus" text
- Timer in centre of screen
- "Reset" & "Modify" buttons
- Play/Pause Button
- Mini motivational message: On the break screen it gives the user healthy reminders such as getting fresh air and drinking water. This encourages healthy habits for the user. *As a possible extra feature, it could randomly display a reminder from a selection of prewritten reminders.*
- Streak counter
- Session counter
- Skip Session button

I have put these elements in the same position as they are in the Focus Screen as it will seem to the user that the screen has not changed (apart from the colour scheme and some text changing). This makes the app feel more simplistic, meeting my [requirements](#). Rather than having a complex app, such as having lots of different screens to try and navigate, my app will feel as though it only has two; The timer screen and the modify screen, despite having four (Focus, Break, Long Break, and Modify Screens).

Break Session screen



Long Break screen:

I have created a long break screen. This screen is shown after four focus sessions have been completed.

I have changed the colour scheme of this screen from the shorter break screen. I have kept the same colours as the break screen but have instead swapped the accent and background colours (the lighter and darker shades of blue). This is so that it can easily be differentiated by the user. Furthermore, I have added text at the top of this screen titling “Long Break” to ensure the user notices the difference between the screens (apart from the timer).

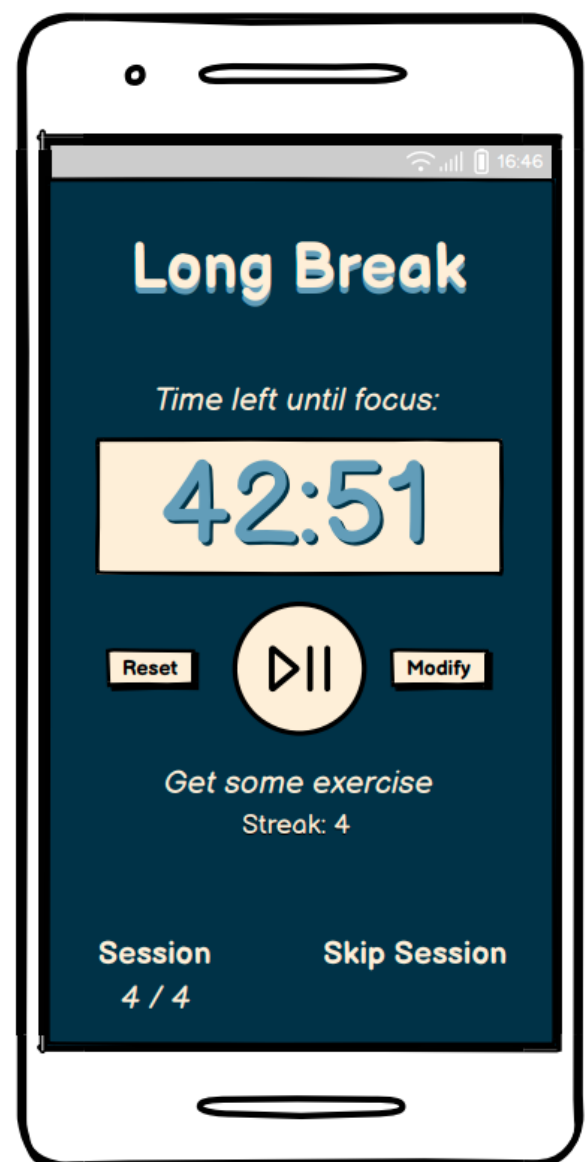
Again, this has the same features and elements as the Focus and Break (short break) screens (*please see Focus & Break Screen*).

Additional notes:

For the titles on each screen and the numbers for the timer, I have added a coloured shadow to make my app more visually appealing ([meeting the requirement](#)).

On top of this, certain text is in bold to stand out to the user (“Session” and “Skip Session”). Other text, “Time left until...” and the mini motivational messages, are in italics to separate them from other elements on screen. This makes the text, and app overall, more interesting to look at as opposed to plain text that all looks the same.

Long Break screen



Modify screen:

I have made this screen to allow the user to adjust the duration of the timers, and the number of sessions that need to be completed before a long break is rewarded (break interval).

This screen is an essential feature of my app, as it allows the user to tailor the app to their preferences. By having this feature in my app, it will encourage the user to use my app more, as it has customisable features that allow them to make my app appeal more to them specifically. This gives my app a more gamified experience and further supports students with their general study skills ([meeting both requirements](#)).

This screen can only be accessed when a timer is paused; This prevents the user from (intentionally or unintentionally) avoiding focusing for the full duration of the timer. To alert the user of this, if they try pressing the Modify button while the timer is running, a notification can be displayed to them to remind them to pause the timer before attempting to modify session durations.

I have mixed the colour schemes of the Focus and Break sessions for this screen as it allows the user to modify the timers of both screens and is therefore related to both colour schemes.

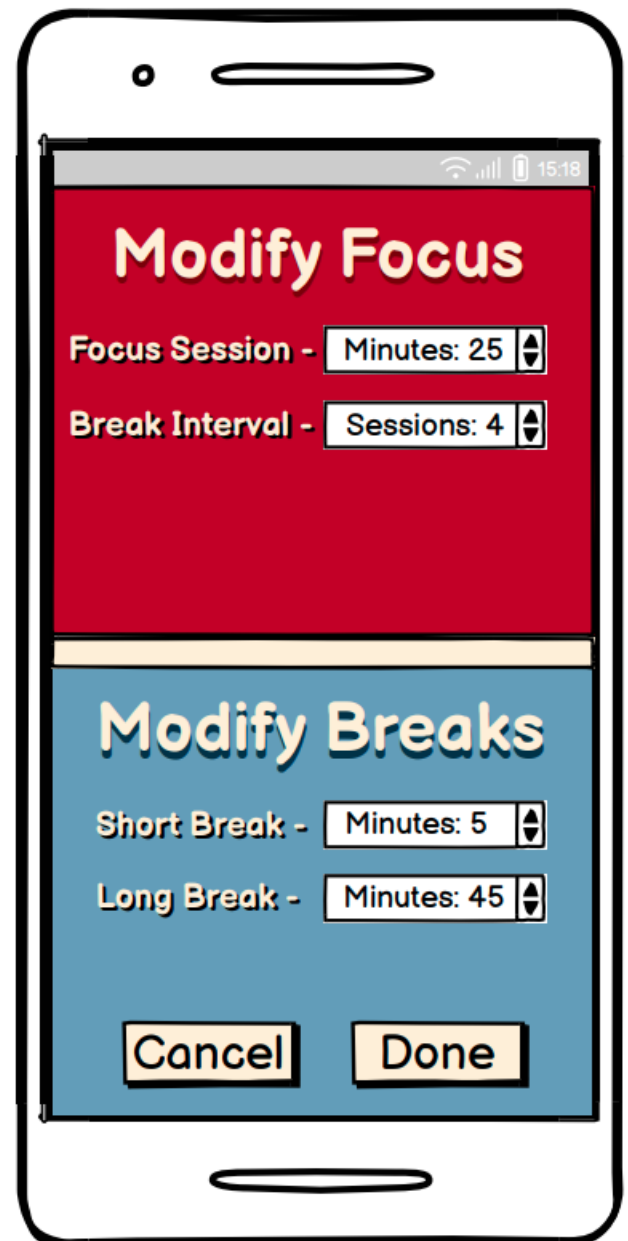
This screen will have buttons that allow the user to increase or decrease the timer durations and the break interval, with the duration and break interval values displayed on screen.

Finally, I have also created two buttons, a Cancel button and a Done button. The cancel button will take the user to the previous screen without changing anything, whereas the done button will update the values to the adjusted values.

Additional Notes:

For the modify screen, I may put a minimum amount of time for each type of session, break, and break interval. This would stop the user from setting focus sessions to a very

Modify screen



short amount of time, such as 1 minute, and the “short breaks” to long amounts of time such as 60 minutes.

If the user were to abuse the modify feature, it would ruin the purpose of this app, as they would no longer be studying & working effectively. By putting in minimum and maximum durations, it prevents the user from misusing my app, ensuring that they study effectively instead.

Cross-platform Suitability:

My design will be suitable for different devices, as it will resize to fit the user’s screen. My app will be displayed in a portrait orientation, as this is most fitting for a mobile phone (which is the most likely device my users would use, as students); However, this would also be suitable for larger mobile devices, like a tablet, as it can also be used in a portrait orientation. This ensures my app to meet the requirement of being able to [run on an Android or iOS mobile device](#).

Feedback Review

For my feedback, I sent my write-up document to two people to review. When I did this, I was not specific in what I wanted my reviews to include and talk about. This led to confusion when evaluating the feedback; However, I still got useful feedback from this.

If I were to redo this part of the project, I would create a simple survey to collect feedback from people; This would ensure I got and clarified what kind of feedback I wanted.

I have explained what has happened alongside the feedback given below:

Feedback Review 1:

For one of my reviewers, who is taking the same course as me, we had an audio call to talk and note down feedback about each other's current mobile app design and write-up at the same time.

During this call I had not specified clearly enough the importance to me of written down notes, leading to most of the feedback being verbal. Because I thought the verbal feedback was also being written down, I did not note them down myself, this led to me forgetting what was said at the time as I was focused more on giving feedback for their app.

Here is the review of some of the notes I got from the first reviewer:

- **Customisation based on streak count** – This was based on the idea of my app's streak feature. This referred to design elements of my app (such as the timer) having either unlockable or a change in designs at certain streak milestones e.g.: streak 5, 10, 20. By adding this feature, it would add to the gamified aspect of my app ([meeting the requirement](#)), enticing students further to maintain a streak as it would give them rewards for achieving milestones; This also gives them specific goals to try and meet rather than just maintaining a streak on its own.
- **Boxes around buttons** – Clear differentiation of what text is a clickable button. This is mainly referring to the "Skip Session" button in my app wireframes as it looks like text (rather than a clickable button). *To guarantee this will not be an issue, when creating my app, I will ensure that all interactable elements of my app have boxes around them to differentiate them from standard text. This will also ensure my app meets the [Intuitive to use requirement](#).*
- **Explain what a Pomodoro timer is** – This feedback informed me that I had not properly explained what a Pomodoro timer is in my write-up. Because of this, it may lead to confusion for those reading my write-up or users who may not have heard of a pomodoro timer before. *To ensure students know how my app works, when I*

develop my app, I would have to create either a very brief explanation when they start the app or a help menu they can access at any time. This would also ensure my app meets the [Intuitive to use requirement](#).

Feedback Review 2:

For my second reviewer, they added Comments to the word document of my write-up; This is again due to lack of clarifying what form I wanted my feedback in. However, this gave me feedback on both the write-up and my app's design when I was looking for feedback about my app's design specifically.

Here is the review of the feedback from this person:

- **Pause Button feature abuse** – One of the points this person talked about was that the user could abuse the pause button element of my app. This would allow them to pause and wait a long time, meaning they could take breaks during focus sessions rather than only pausing in an emergency or for an important reason. They could also pause during a break session, allowing them to technically extend the period of their break.

By abusing the pause button, it would ruin the purpose of my app. Because of these reasons, they suggested to remove this feature; However, I believe it is a vital feature of my app in the case that the user must deal with something more important or an emergency, in which case they do not deserve to lose their streak. If they lost their streak in this scenario, it would be unfair and could therefore put them off using my app.

To prevent these two issues, I could set a time limit to the pause (to prevent the user from having an unlimited break) and a pause cooldown (to prevent the user from pausing it every time the pause timer ends). However, by having a limit to the pause button, if someone had an emergency over the duration of it, it would continue running focus session after focus session over time until they closed the app.

Furthermore, by adding these features, it could take away from the independence the user has, which is a key [requirement](#) of my app. The purpose of my app is to help assist students with their independent studying, this requires that they put in some effort themselves and don't try and abuse the app entirely.

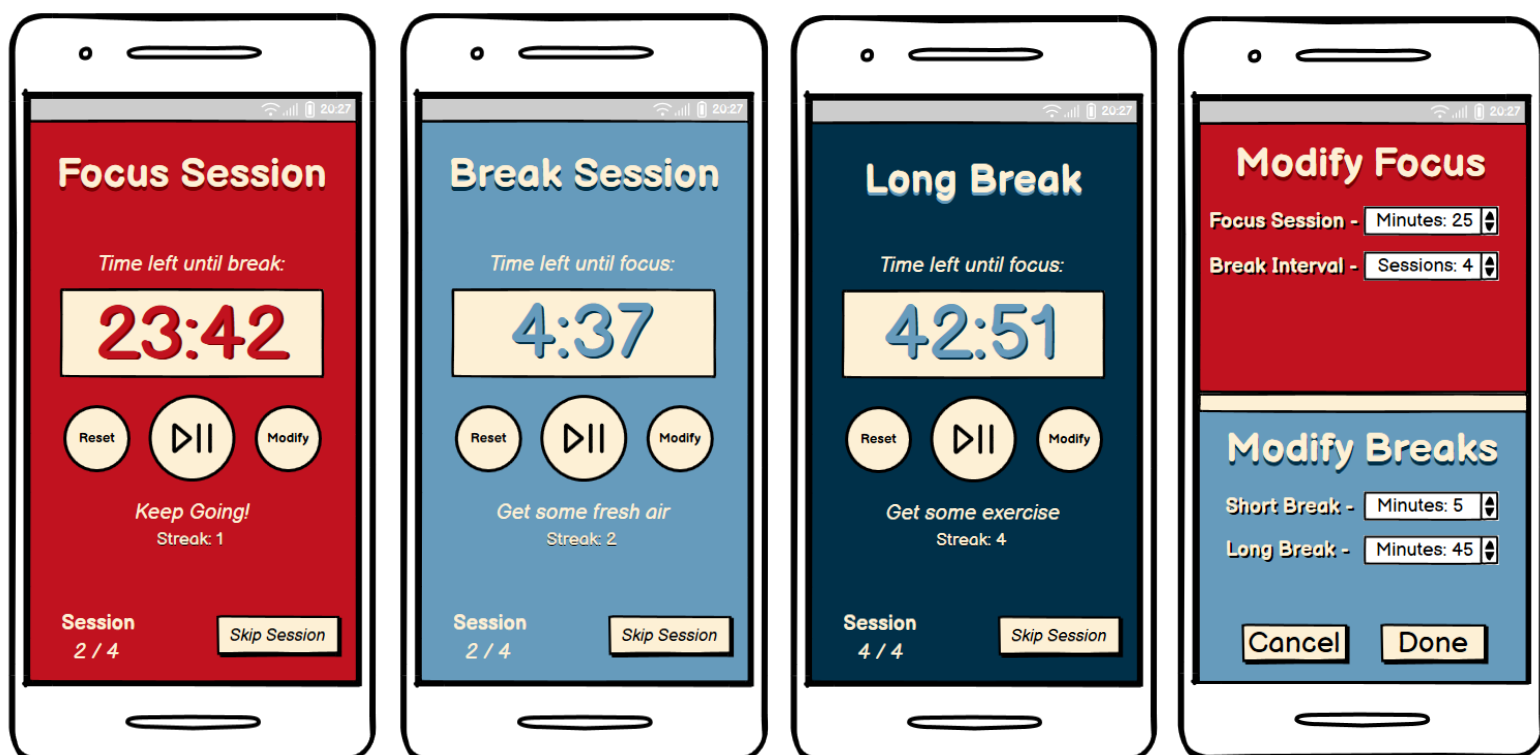
If I were to implement these features, the user would be able to just ignore the timer entirely regardless of whether it has started or ended. Because of this, I am not going to add these features.

- **Access to the modify screen** – Another point this person made was that users should not be able to access the modify screen halfway through a timer, and instead only be able to access it at the start of a session. I have currently got it written up as

only being able to access the modify screen while the timer is paused, however, as they would be able to just reset their timer manually and then use the modify screen, I am not going to change this feature. However, I will need to ensure that the timer is reset when the user sets a new time and or clicks “Done”, whereas it will continue from where they left off if they click “Cancel” on the modify screen.

Refined Design (Wireframes)

I have refined my original Balsamiq Wireframes design based on the feedback I got from potential users of my app:



Based on feedback, to ensure a clear differentiation between what pieces of text are buttons (as opposed to just text), I have made the Reset and Modify buttons larger and rounded in the same shape as the Play/Pause button. This also allows the user to press the buttons easier as before they were a small rectangular shape, which would be difficult to press on a mobile device comfortably.

Furthermore, I have added a clear border around the Skip Session button with a coloured background to make it stand out clearly as a button. With this updated design, it will be very clear to the user what elements are buttons, and which are just text.

These adjustments ensure my app meets the [intuitive to use requirement](#).

I will now review my current design before going on to create it.

App Design Justification & Review

As part of the [assignment brief](#), it states several requirements to ensure my app can support students with their study skills and independence. I will now go into detail on how my app will meet each of these requirements and [my own](#):

Requirement 1 – *Must help support students with general study skills (non-subject specific):*

My app supports students with managing their time effectively. It does this by splitting up their work/study sessions into bursts of studying followed by short breaks. This ensures that the students who use my app will spend the majority of their time working, while still taking healthy breaks to retain their focus.

As my app works as a timer for these sessions for the user, the user can decide what to do during the sessions; Because of this, my app assists them with *general* study skills, as opposed to *subject specific* study skills.

Requirement 2 – *Must utilise gamification to help motivate students to engage with my app:*

My app will utilise gamification to help motivate students to engage with my app and work effectively towards their studies. One way it does this is by having a streak timer which is incremented each time the user completes a focus session or is reset when a user skips a focus session (see [Requirements](#)).

By having a streak feature, it will entice students to work for the full duration of the focus session to avoid losing their 'streak' and complete more focus sessions in the future to further increase their streak. I have added this to my app as it will keep students engaged and draw them back to studying independently.

Furthermore, my app has a motivational message (user praise) that encourages the user to continue studying (see [here](#)).

Requirement 3 – *Must be visually appealing:*

To make my app design visually appealing, I have used a popular colour scheme and simplistic, but still appealing, fonts (see [App Theme & Style](#)).

I have also designed my app with a simplistic layout in mind to avoid overwhelming the user, this makes for an appealing app as it only has useful and important information on screen (see [Layout](#)). Furthermore, I have used shades of colours that are not too bright or dark to ensure the design is not too flashy and doesn't stand out harshly.

Finally, I have made use of colour psychology related to the connotations of colours. *For more detail about this please see [Use of Colour Connotations](#) and [App Theme & Style](#).*

Requirement 4 – *Must be intuitive to use and not require additional guidance:*

To make my app intuitive to use, I have made use of predictable element placement, such as pause buttons beneath timers, titles at the top middle, and buttons near other elements they relate to. I have also used recognisable elements that are commonly used in other apps, such as buttons and counters. Because of this, users will know how to interact with them or what they mean (see [Layout](#)).

By using familiar elements, in predictable places, it reduces the chance of the user getting confused on how to use my app. Furthermore, my app has minimal navigation, further reducing the chance of students getting confused on how to use and traverse my app. The app navigation includes clicking a "Modify" button to take the user to a modify sessions screen (see [Layout](#)), and to traverse back from the Modify screen the user is offered a "Cancel" or "Done" button, both of which are common types of button in apps and (mobile) devices. The buttons in my app design are also all labelled, making it clear on what they do, again meaning the user won't need additional guidance.

Requirement 5 – *Must run on an Android or iOS mobile device:*

I am going to build my app using [Thunkable](#). Thunkable allows for the creation of Android and iOS apps, however, as I own an Android phone I am going to create an Android app; Furthermore, this means I can easily test it on the target platform using Thunkable's live app testing feature.

My App's Development

App Test Plan

Below I have written a test plan, I will test my app against this plan when it is completed to ensure that all features work as intended.

Functionality Test Plan:

Test	Description	How I will test it	Expected Result	Actual Result
Focus Timer	Test that the Focus session timer functions correctly.	Press the play button and watch the timer from start to finish.	Timer counts down from the full duration and stops at zero.	N/A
Break Timer	Test that the Break session timer functions correctly.	Press the play button and watch the timer from start to finish.	Timer counts down from the full duration and stops at zero.	N/A
Long Break Timer	Test that the Long Break session timer functions correctly.	Press the play button and watch the timer from start to finish.	Timer counts down from the full duration and stops at zero.	N/A
Play/Pause Button	Test that the play/pause button works correctly.	While a timer is running, click the button and see response.	Timer pauses & resumes accordingly across all session timers.	N/A
Alarm/Notification	Alarm/notification that notifies the user that the current timer has ended.	Let a timer run until it is finished and see if a notification occurs.	App notifies the user that the current timer is finished.	N/A
Adjustable Focus Timer	User can update the duration of the focus timer.	Attempt to update the timer and view the session timer.	Timer updated correctly.	N/A
Adjustable Break Timer	User can update the duration of the break timer.	Attempt to update the timer and view the session timer.	Timer updated correctly.	N/A
Adjustable Long Break Timer	User can update the duration of the long break timer.	Attempt to update the timer and view the session timer.	Timer updated correctly.	N/A

Test	Description	How I will test it	Expected Result	Actual Result
Session Change	The change between screens after a timer has ended.	Wait for each timer to run out.	Change between screens when the timer runs out.	N/A
Long Break Functionality	Long break opens correctly.	Finish the amount of focus sessions needed to be rewarded long break.	Change screen to the focus timer when the amount of focus sessions is correct.	N/A
Reset Button	Resets the current timer.	Click the reset button while the timer is paused and running.	Timer duration resets (and pauses?)	N/A
Session Counter	Counts the number of focus sessions out of a total.	Complete focus sessions and watch the counter.	Counter is incremented for each focus session completed, and resets after long break.	N/A
Skip Session Button	Allows the user to skip the current session.	Click the skip button and see if the screen changes.	Screen changes to the next appropriate session. This should also reset the streak.	N/A
Modify Button	Allows the user to modify timers/settings	Click the button and see if it takes me to the modify screen.	Screen changes to the modify screen.	N/A
Streak Counter	Keeps count of the number of focus sessions done without skipping.	Complete focus sessions.	Streak counter should be incremented when focus sessions completed. Should also save the streak data when app is closes.	N/A
Cancel Button	Cancels any changes.	Click cancel button.	No changes made, user returned to current session.	N/A
Done Button	Applies any changes.	Click done button.	Changes applied, current timer reset and user returned to current session.	N/A
Break Interval Modification	Allows the user to adjust the long break interval.	Edit the interval and complete that number of focus sessions.	Long break session started when interval reached.	N/A

Usability Test Plan:

Test	Description	How I will test it	Expected Result	Actual Result
UI Readability	Readable UI.	Have several people (including me) look at my app and give feedback on this.	All text is clear readable.	N/A
Clear & Simplistic Layout	Layout being uncluttered or overwhelming.	Have several people (including me) look at my app and give feedback on this.	UI and design are not overwhelming to the user.	N/A
Intuitive to use	App being easy to use without having to explain everything.	Have several people (including me) look at my app and give feedback on this.	App is intuitive to use, and users understand what each feature does.	N/A

Stability & Performance Test Plan:

Test	Description	How I will test it	Expected Result	Actual Result
Crash & Stress Test	Testing if the app crashes with normal and excessive use.	Use the app normally to test it crashing. Use the app & its elements excessively to stress test.	App does not crash.	N/A
Performance Test	Testing if the app slows down in running with normal and excessive use.	Use the app normally to test it dropping in performance. Use the app & its elements excessively to test it dropping in performance.	App does not drop in performance.	N/A

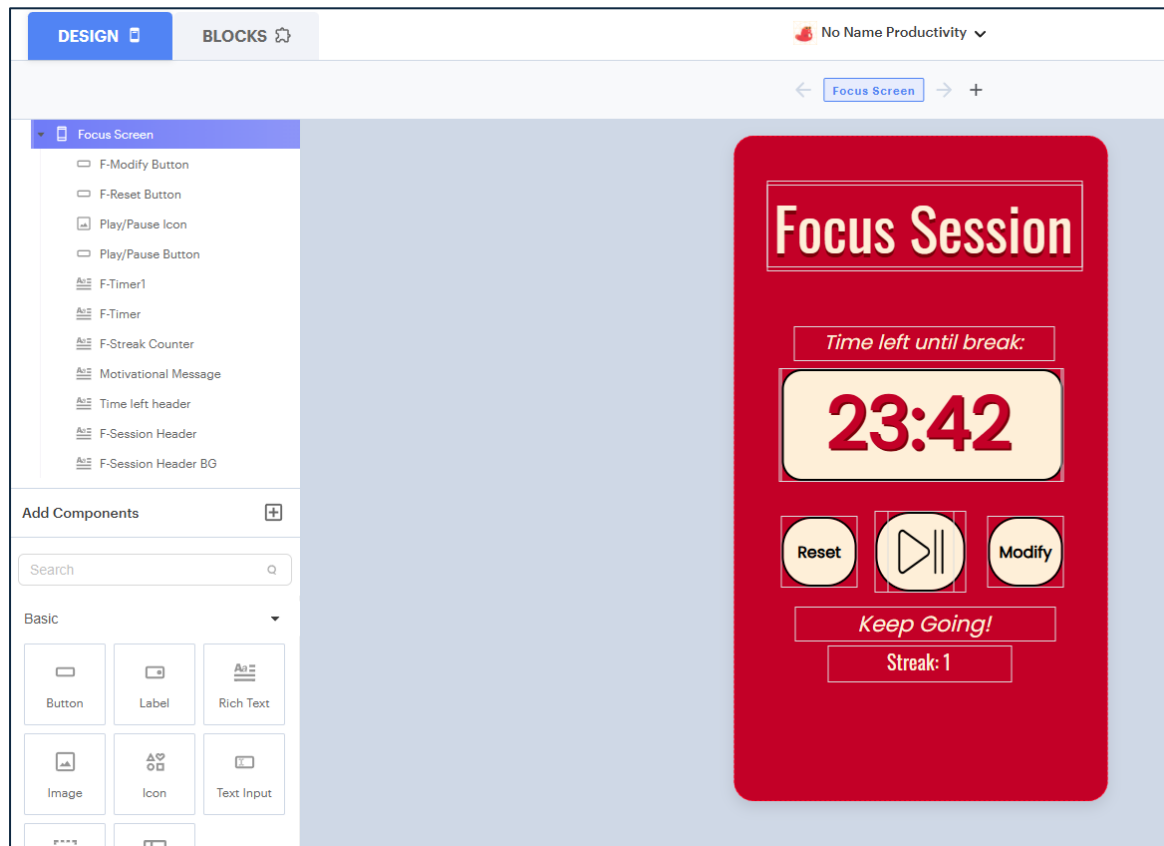
Creating the App

To create my app, as previous mentioned, I will be using [Thunkable](#). This is an in browser mobile app builder. I have chosen to use this development environment as I am familiar with it, having created an app with it before, and because it is a drag and drop app builder; This means that I will be able to create my app quicker than if I were to use software I was unfamiliar with or to code it line by line.

As I have a limited amount of time on this project, I do not have time available to learn how to use other software or code it line by line; Again, this is why I have chosen to create my app using Thunkable.

17/03/2026

Started developing the Focus screen:



Today I created the main design of the Focus screen of my app and started writing some of the code. I have decided not to use the Pacifico font I chose in my design (see [here](#)) as it could be quite hard to read for certain users; However, I have still chosen to use the Oswald and Poppins fonts.

Timer Display issue:

To display the timer correctly, I had to research how to display timers in the correct format. I used this thread: <https://community.thunkable.com/t/how-to-display-timer-in-hhss-format/3128139>

Play/Pause Icon issue:

Originally for the icon of the play/pause button, I used images from the internet:

<https://thenounproject.com/browse/icons/term/pause-play-button/>

<https://creazilla.com/media/emoji/44566/play-or-pause-button>

However, because I didn't like the look of the images I found, and the icon would have to be placed above the button (leading to problems when clicking it), I looked into an alternative solution.

The solution I found was to edit the button's text to what looked like a pause symbol, for when the timer was running, and then switch to text that looked like a play symbol when the timer was paused. To find how to make symbols with a keyboard, I used this post: <https://www.facebook.com/leptieltsteacherfrank/posts/how-to-make-symbols-with-keyboardalt-0153-trademark-symbolalt-0169-copyright-sym/1117745803700981/>

This explained how to use the "Character Map" software that comes with computers, I had not known about this before.

Repeated Code issue – Colour:

To reduce the amount of repeated code in my app, I created variables that stored the colours in my app's colour scheme (dark red, light red, off white, light blue & dark blue) so that I could use them to set certain text to the right colour when updated.

To do this I needed the RGB values, however I only had the hex values. TO convert these, I used this website: <https://htmlcolorcodes.com/hex-to-rgb/>

18/03/2026:

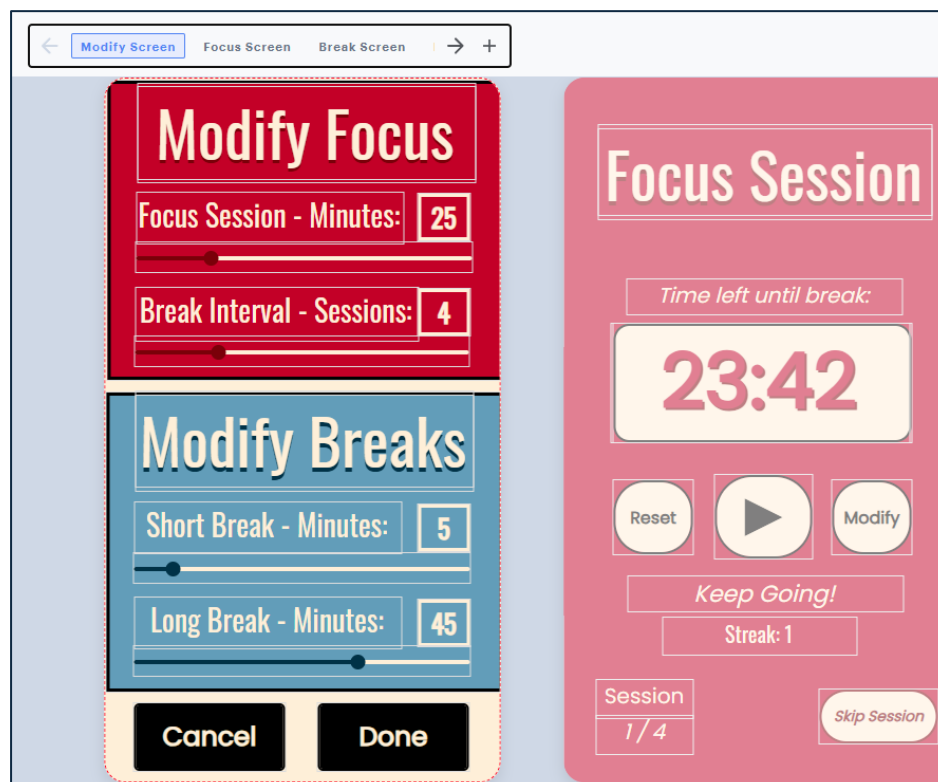
I set up the modify screen. I changed the layout to my original design so that the Cancel and Done buttons were clearly relating to all settings (rather than them both being in front of the same background as Modify Breaks) and stood out.

I also swapped the duration up and down buttons for a slider. I changed it to this as it is more intuitive and easier to use, especially on a mobile device, than small buttons. These sliders update temporary variables accordingly and are either saved, and update the main variables when Done is pressed, or it does not update the main variables when Cancel is pressed. Both buttons then take the user back to their previous screen. To do this, I have a variable storing the previous screen's name, which is then fed into code to ensure the user returns to their previous screen/session.

Furthermore, these sliders can have minimum and maximum values, giving them a range of the length of the sessions: For Focus, the range is 15-60 minutes; For the Break Interval, the range is 2-10 sessions; For the Short Break, the range is 3-20 minutes; For the Long Break, the range is 25-60 minutes. The slider defaults are seen in the image below (25, 4, 5, 45, respectively). I have set limits and maximum to ensure the user doesn't only work for too short amount of time and doesn't have too long of a break.

To test my app as I develop it, I will set the minimum for these sliders to 6 seconds, this will allow me to simulate a timer ending over time without having to wait a long time.

Also added more to the focus session screen including session counter and skip session button and fixed an issue with the pause button.

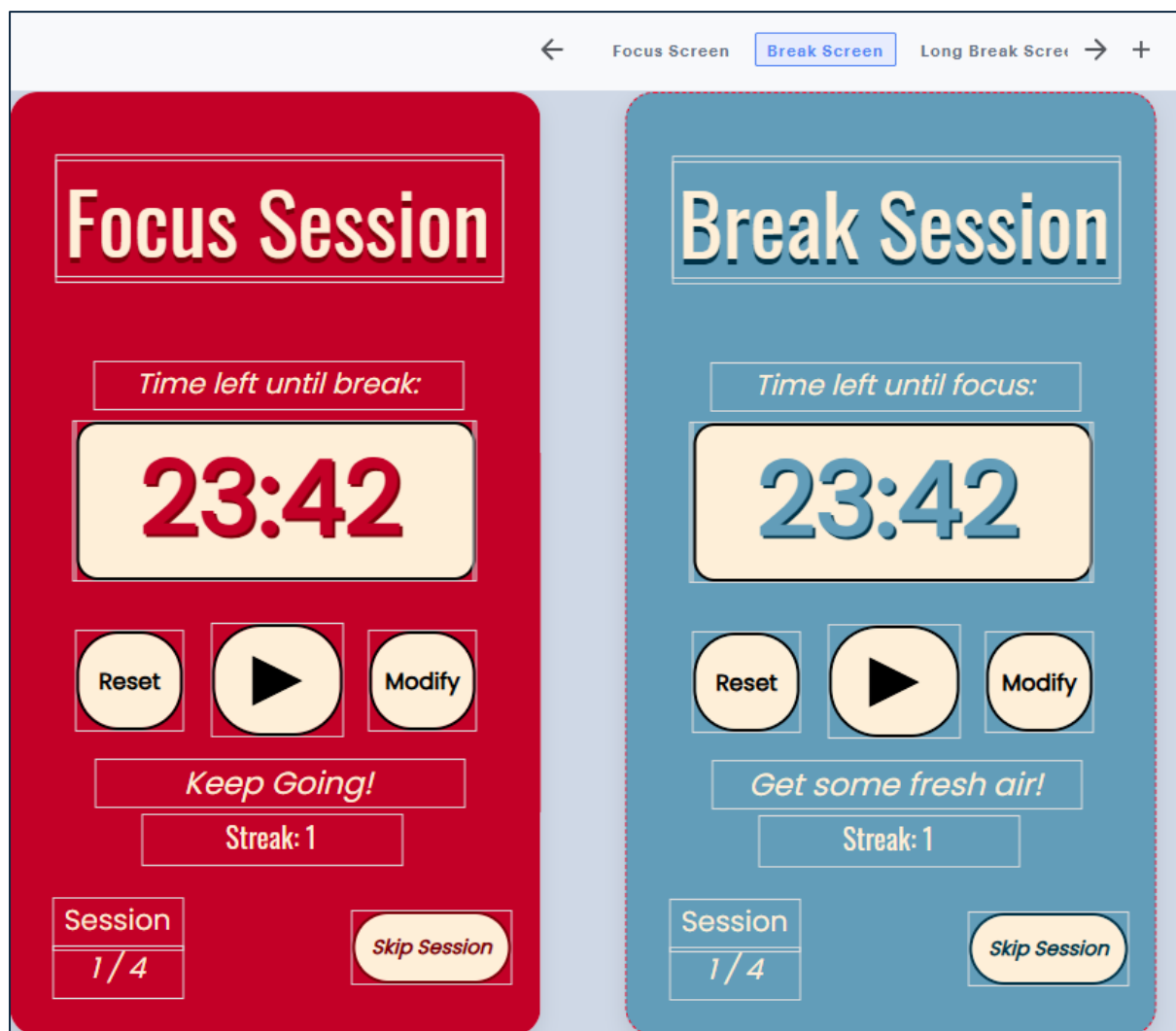


19/03/2026:

Finished the focus session screen code (excluding session & streak counter and skip session button).

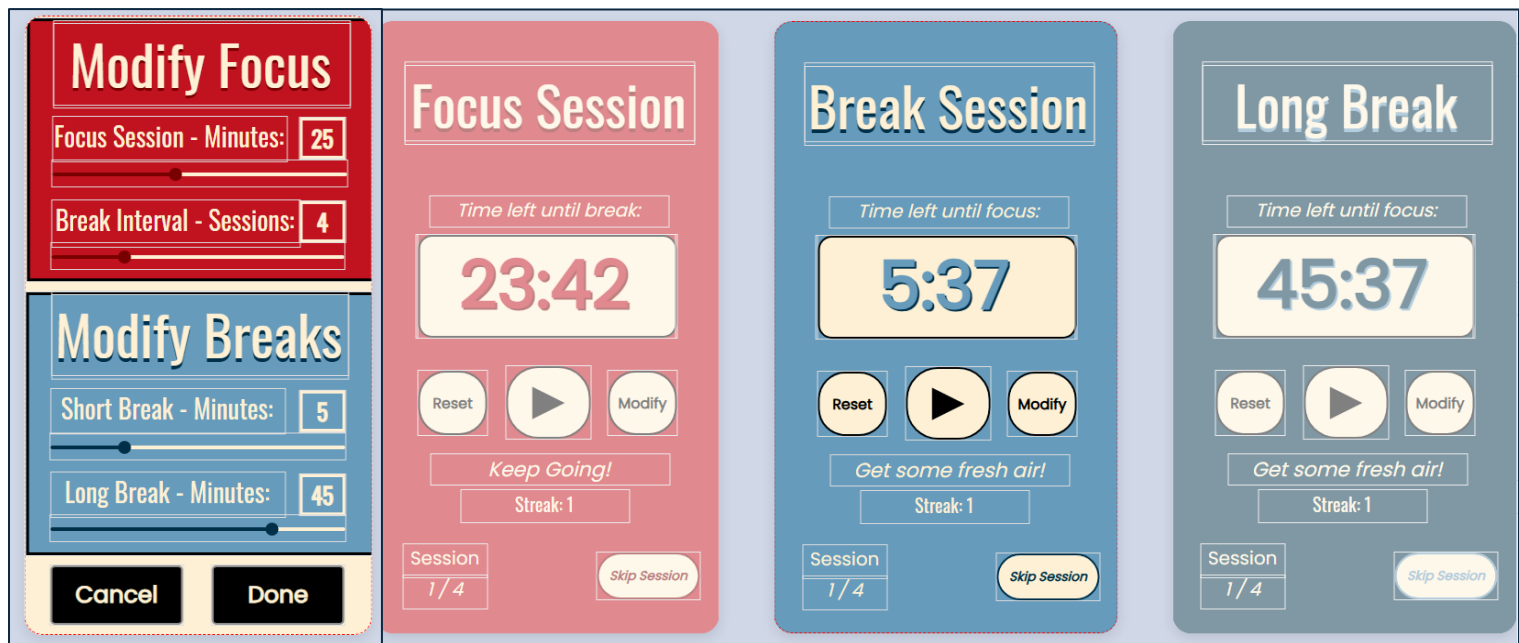
Created break session screen including code (also excluding session & streak counter and skip session button) this screen uses the same code as focus session, so it did not take much time at all.

Tomorrow I will aim to finish the long break session and the session counters.



20/03/2026:

Screens so far:



Bugs:

- Reset button after timer ended – This allowed the user to reset the timer after it had already ended, this causes issues with the app.
- Focus session timer occasionally counts down quicker (e.g. in 2s, 3s, 4s) – I am unsure what causes this issue. See MP4 files **Countdown Bug** and **Countdown Bug 2**.
- Using modify screen after focus timer is done resets the timer but still awards session completed – This is a similar issue to the reset button; To fix both issues, I need to add code that disables them when the timer finishes. See MP4 file **Modify reset countdown bug**.

The “Forever” loop issue:

Forever loops if get stuck can cause crashes instantly. By having Forever loops in my app, it has started to cause my app to crash and freeze. See MP4 file **Thunkable Freezing - Potentially loop bug**.

What I did today:

I implemented to session counter feature to my app. As I was doing this, I encounter a lot of bugs and issues (*discussed above*), because of this, I then had to spend my time today trying to resolve the bugs and issues.

Today I also found and fixed the mentioned bugs by removing “Forever” loops and replaced with timers (functions that are called every set interval e.g.: 1 second).

I have fixed the focus session screen and ensured buttons are activated and deactivated appropriately; the Modify button can only be pressed when the timer is paused and not finished and the Reset button can only be pressed when the timer is not finished (this button will also pause the timer after resetting it).

When the timer finishes, I have written code that will change the Play/Pause button to a “Next” button; When the user presses this, it will take them to the next appropriate session (Short Break or Long Break, dependent on if the session interval has been met for a long break)

Furthermore, the Play/Pause button is disabled when there are 3 seconds left on the timer. This is so the user does not spam the Play/Pause button as the timer ends, as this would otherwise cause a bug for the button to never change to a “Next” button, softlocking the user as all buttons would be deactivated.

Finally, to make my code easier to read, and more efficient, I created and used functions for repeated code.

Tomorrow's Goals:

Tomorrow I will implement the Skip Session button on the focus screen.

Then I will replace the existing code for the Break and Long Break screens with this new updated code that does not use Forever loops. This should fix any remaining bugs. On top of this I will implement the same Skip Session button code for the other screens.

21/03/2026:

Today's Goals:

Finish developing the app before review & optimisation.

- Double check there are no Forever loops left in Focus and Modify screens
- Copy and modify code accordingly from Focus screen to the Break and Long Break screens
- Confirm that session counter is working still
- Implement the streak feature
- Implement the Skip Session button – An additional note is this should only reset the streak when it skips a Focus session, not a break session.
- Implement a notification or alarm SFX when the timers end

What I did today & Bug log:

Fixed all outstanding bugs, copied modified code (and adjusted it) to the two break screens, ensured the session counter worked, implemented the streak feature & skip button, and an on-screen alert for when the timers end.

In order to add alerts, I had to research how to code them in Thunkable. To do so, I watched their tutorial: <https://docs.thunkable.com/blocks/app-features/alert>

Today's progress on my app took a long time to develop as I encountered a lot of bugs. I have gone into more detail about some of these below:

- **Infinite Loop issue/bug** – Originally, I created my app using infinite loops, this led to my app freezing and crashing as soon as it started. This is because they looped repeatedly at a high speed and locked my app's resources. To resolve this issue, I deleted all infinite loops and replaced them with Thunkable's Timer Component. This fixed the issues I had with the infinite loops.
- **Button Update bugs** – On the focus screen, the Modify and Reset buttons were stuck in a disabled state when they should have been clickable. The issue that caused this was a background timer, that I had set to update buttons, was running too often (every millisecond). To fix this, I removed the timer and created a function that updated the on-screen buttons; This function was then called every time a button's state may have changed. By doing this, it also fixed the issue from yesterday where the user was able to reset or modify the timer after it had ended.

- **UI Update bugs** – On the break screen, the streak counter was not updating correctly and showed different values to the streak counter on the focus screen. This took a while for me to resolve as I could not find any code that would have changed the value of the variable to cause this. Eventually, I realised that I had not written the code to update the UI yet, therefore it was displaying the default value. By writing the correct code, my app then functioned correctly.

Conclusion:

I have now finished developing my app and am going to move onto testing it.

Testing the App Functionality

To test my app, I am going to run the tests I planned [here](#). Please see the results below:

Functionality Tests:

Test	Description	Expected Result	Actual Result
Focus Timer	Test that the Focus session timer functions correctly.	Timer counts down from the full duration and stops at zero.	<i>Functions Correctly</i>
Break Timer	Test that the Break session timer functions correctly.	Timer counts down from the full duration and stops at zero.	<i>Functions Correctly</i>
Long Break Timer	Test that the Long Break session timer functions correctly.	Timer counts down from the full duration and stops at zero.	<i>Functions Correctly</i>
Play/Pause Button	Test that the play/pause button works correctly.	Timer pauses & resumes accordingly across all session timers.	<i>Functions Correctly</i>
Alarm/Notification	Alarm/notification that notifies the user that the current timer has ended.	App notifies the user that the current timer is finished.	<i>Functions Correctly It displays an on-screen alert.</i>
Adjustable Focus Timer	User can update the duration of the focus timer.	Timer updated correctly.	<i>Functions Correctly Users can use a modify screen</i>
Adjustable Break Timer	User can update the duration of the break timer.	Timer updated correctly.	<i>Functions Correctly Users can use a modify screen</i>
Adjustable Long Break Timer	User can update the duration of the long break timer.	Timer updated correctly.	<i>Functions Correctly Users can use a modify screen</i>

Test	Description	Expected Result	Actual Result
Session Change	The change between screens after a timer has ended.	Change between screens when the timer runs out.	<i>Functions Correctly</i>
Long Break Functionality	Long break opens correctly.	Change screen to the focus timer when the amount of focus sessions is correct.	<i>Functions Correctly</i>
Reset Button	Resets the current timer.	Timer duration resets (and pauses?)	<i>Functions Correctly And pauses</i>
Session Counter	Counts the number of focus sessions out of a total.	Counter is incremented for each focus session completed, and resets after long break.	<i>Functions Correctly</i>
Skip Session Button	Allows the user to skip the current session.	Screen changes to the next appropriate session. This should also reset the streak.	<i>Functions Correctly And resets streak</i>
Modify Button	Allows the user to modify timers/settings	Screen changes to the modify screen.	<i>Functions Correctly</i>
Streak Counter	Keeps count of the number of focus sessions done without skipping.	Streak counter should be incremented when focus sessions completed. Should also save the streak data when app is closes.	<i>Functions Correctly Saves the streak data to the phone itself. Stores the variables to phone.</i>
Cancel Button	Cancels any changes.	No changes made, user returned to current session.	<i>Functions Correctly</i>
Done Button	Applies any changes.	Changes applied, current timer reset and user returned to current session.	<i>Functions Correctly</i>
Break Interval Modification	Allows the user to adjust the long break interval.	Long break session started when interval reached.	<i>Functions Correctly</i>

Notes:

My app fully works as I originally intended it to. It successfully passed all the functionality test criteria. Please see MP4 file: “**No Name Productivity - Version 1.0 Screen Recording - 2026-03-21**” for overall test and app function (*please note I have adjusted the modify screen’s slider minimum values to allow a simulation of a timer running out quickly without being skipped*).

I also only used two of my initial choices for fonts (Oswald and Poppins) as they were the clearest to read.

Developed App Feedback Review

After creating the first version of my app, I got several people to test and review my app. These testers gave feedback on improvements I could add to my app in the optimised version.

I have gone into more detail on these reviews below:

Reviewer 1:

Improvements:

Layout Adjustment – To improve my app, this person suggested that I adjust the layout of my app (for the focus, break, and long break screens) so that it is more user friendly. They stated that it would be easier for the user to press the Play/Pause, Reset, and Modify buttons if they were positioned closer to the bottom of the screen.

I will implement this in the optimised version of my app; To do this, I will swap the three mentioned above buttons with the motivational message and streak counter.

Skip Session Confirmation – When this person tested my app, they accidentally pressed the skip session button. This is an issue, as if a user accidentally pressed the skip button, it would reset their streak which they could have spent a long time working towards; This may put people off continuing to use my app due to frustration/upset.

I will resolve this issue in the optimised version of my app; To do this, I will have an on-screen alert pop-up that allows the user to confirm or cancel the skip session button. This will ensure they won't accidentally press the skip button.

Reviewer 2:

Improvements:

Streak highscore – Having a ‘highscore’ for the user’s highest streak. This would encourage the user to try and beat their old streak (if they lost it), further gamifying my app. I also had this idea when I was developing my app.

I will implement this feature by having another stored (to the user’s device) variable that stores a streak highscore. This would be updated correctly and never reset (unless the user deleted their stored data on their phone). I will also need to display it on the screen, potentially at the top so it is out of the way.

Vibrate device at the end of a timer – By making the user’s device vibrate when a timer finishes, it will make them more likely to notice the current timer has ended. Currently my app only has an on-screen alert pop-up.

I will implement this feature by adding in built-in code (Thunkable has a function that does this), to where a timer ends.

Extra Ideas:

While developing my app, I came up with this idea as a possible future addition:

Timer Font rewards – The font of timers changing as the user’s streak reaches certain milestones; This would add to the gamification of my app and encourage app use.

To do this, I would swap out where a font is set in the code, for a variable which would store a string of a font’s name. This string could then be updated when streak milestones are met, which would in turn update the on-screen timer’s text to the font. Font size may also need to be adjusted; However, this could also be stored as a variable which is updated based on the streak milestones.

I am not going to prioritise implementing this, as I want to ensure the features testers suggested are implemented as they are more important to my app’s functionality.

Tests

To fill out & test my app against the usability, stability, & performance test plans, I will get users (who gave feedback on my developed app) to test out my app. The results from this will then be used for the tests.

Usability Tests:

These tests test my app against the brief requirement of being intuitive to use without requiring additional guidance.

Test	Description	Expected Result	Actual Result
UI Readability	Readable UI.	All text is clear readable.	<i>As expected, all text is clear to read and understand.</i>
Clear & Simplistic Layout	Layout being uncluttered or overwhelming.	UI and design are not overwhelming to the user.	<i>As expected, all UI & Design is not overwhelming or cluttered.</i>
Intuitive to use (Brief requirement)	App being easy to use without having to explain everything.	App is intuitive to use, and users understand what each feature does.	<i>As expected, all elements of my app are self-explanatory and intuitive to use.</i>

Stability & Performance Tests:

These tests test my app against the brief requirement of running on an Android or iOS mobile device and my stability & performance itself.

Test	Description	Expected Result	Actual Result
Crash & Stress Test	Testing if the app crashes with normal and excessive use.	App does not crash.	<i>My app eventually stopped working correctly, this caused my app to softlock the user unless they were to restart the app. See MP4 files Stress Test SR part1&2. However, it did not crash.</i>
Performance Test	Testing if the app slows down in running with normal and excessive use.	App does not drop in performance.	<i>As expected, as far as me and beta testers are aware, the app does not noticeably drop in performance at any time.</i>

Developed App – Version 1 Review

As mentioned earlier (see [here](#)), as part of the [assignment brief](#), it states several requirements to ensure my app can support students with their study skills and independence. On top of this, I made my own [requirements](#).

Here are the requirements, and how far I've met them with the first version of my app, below:

Requirement 1 – *Must help support students with general study skills (non-subject specific):*

My app successfully supports students with managing their time effectively. It does this by splitting up their work/study sessions into bursts of studying followed by short breaks (using a pomodoro style technique). This ensures that the students who use my app will spend an effective amount of time working, while still taking healthy breaks to retain their focus and work efficiently.

As my app works as a timer for these sessions for the user, the user can decide what to do during the sessions; Because of this, my app assists them with *general* study skills, as opposed to *subject specific* study skills, hence meeting this requirement.

To further improve on how my app helps students with their general study skills, I would like to utilise the mobile device's vibrate feature to alert the user when the timer ends. This was [suggested by the second reviewer](#) of my app and is something I will add in the optimised version of my app.

Requirement 2 – *Must utilise gamification to help motivate students to engage with my app:*

My app utilises gamification to help motivate students to engage with my app and work effectively towards their studies. One of the ways it does this is by having a streak timer which increases each time the user completes a focus session or is reset when a user skips a focus session (see [Requirements](#)).

By having a streak feature, it entices students to work for the full duration of the focus session to avoid losing their 'streak' and complete more focus sessions in the future to further increase their streak. I have successfully implemented this to my app as planned. Furthermore, my app has a motivational message (user praise) that encourages the user to continue studying (see [here](#)).

To further improve how my app utilises gamification, I would like to implement a streak highscore feature ([suggested by reviewer 2](#)). This would further motivate my app's users

(students) to engage with my app. To ensure that I fully meet this requirement, I will implement this feature in the optimised version of my app.

Requirement 3 – Must be visually appealing:

To make my app design visually appealing, I have used a popular colour scheme and simplistic, but still appealing, fonts (see [App Theme & Style](#)). This makes my app visually appealing as it won't overwhelm the user with inappropriate fonts and uses a colour scheme that people like the look of.

I have also designed & created my app with a simplistic layout in mind to avoid overwhelming the user; This makes for an appealing app as it only has useful and important information on screen (see [Layout](#)). Furthermore, I have used shades of colours that aren't too bright or dark to ensure the design isn't too flashy and doesn't stand out harshly.

Finally, I have made use of colour psychology related to the connotations of colours. *For more detail about this please see [Use of Colour Connotations](#) and [App Theme & Style](#).*

Requirement 4 – Must be intuitive to use and not require additional guidance:

To make my app intuitive to use, I have made use of predictable element placement, such as pause buttons beneath timers, titles at the top middle, and buttons near other elements they relate to. I have also used recognisable elements that are commonly used in other apps, such as buttons and counters. Because of this, users will know how to interact with them or what they mean (see [Layout](#)).

By using familiar elements, in predictable places, it reduces the chance of the user getting confused on how to use my app.

Furthermore, my app has minimal navigation, further reducing the chance of students getting confused on how to use and traverse my app. The app navigation includes clicking a "Modify" button to take the user to a modify sessions screen (see [Layout](#)), and to traverse back from the Modify screen the user is offered a "Cancel" or "Done" button, both of which are common types of buttons in apps and (mobile) devices. The buttons in my app design are also all distinctly labelled, making it clear on what they do, again meaning the user won't need additional guidance.

To further improve the intuitiveness of my app, I will implement two features/adjustments [suggested by the first reviewer](#). This includes adding a Skip Session Confirmation screen; This is a pop-up that appears when the user presses the 'Skip Session' button and will ask the user to confirm whether or not they want to skip the current session. I will also adjust the current layout of my app to ensure that it meets

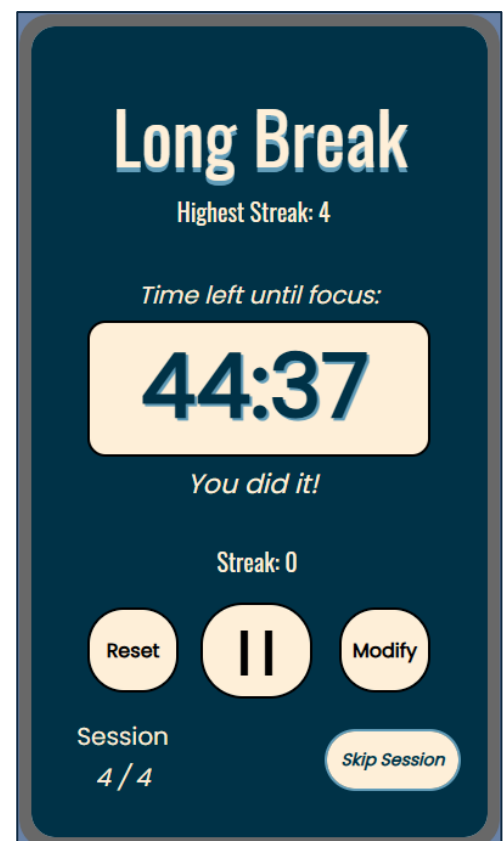
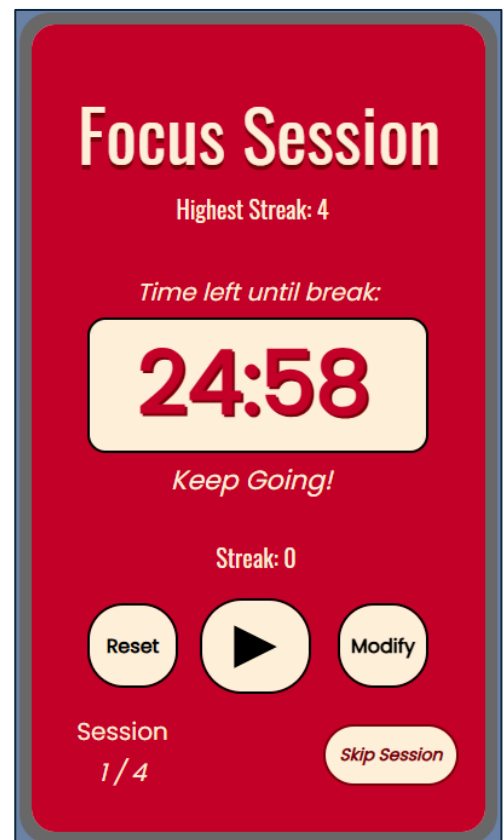
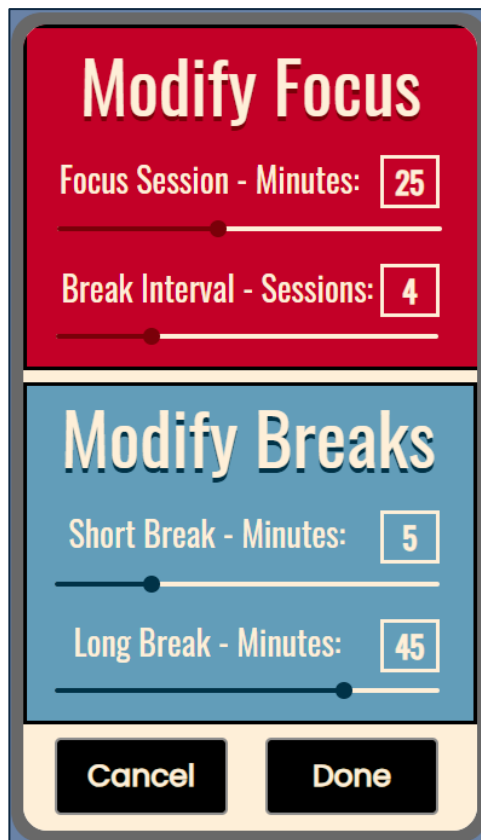
the requirements of being intuitive to user and not require additional guidance; The reviewer suggested that I move the placement of the buttons to be further towards the bottom of the screen as this would allow for a more natural (& intuitive) pressing of these, especially on a mobile device (it would also allow easier use of the app if the user was to hold their phone with one hand). I will implement both of these features in the optimised version of my app to ensure I meet this requirement.

Requirement 5 – Must run on an Android or iOS mobile device:

I have built my app using [Thunkable](#). Thunkable allows for the creation of Android and iOS apps, however, as I own an Android phone I have created an Android app; Furthermore, this allowed me to easily test it on the target platform using Thunkable's live app testing feature (emulating my app on an Android mobile device).

My Optimised App Development

Final Screen designs / layout:



Optimisations

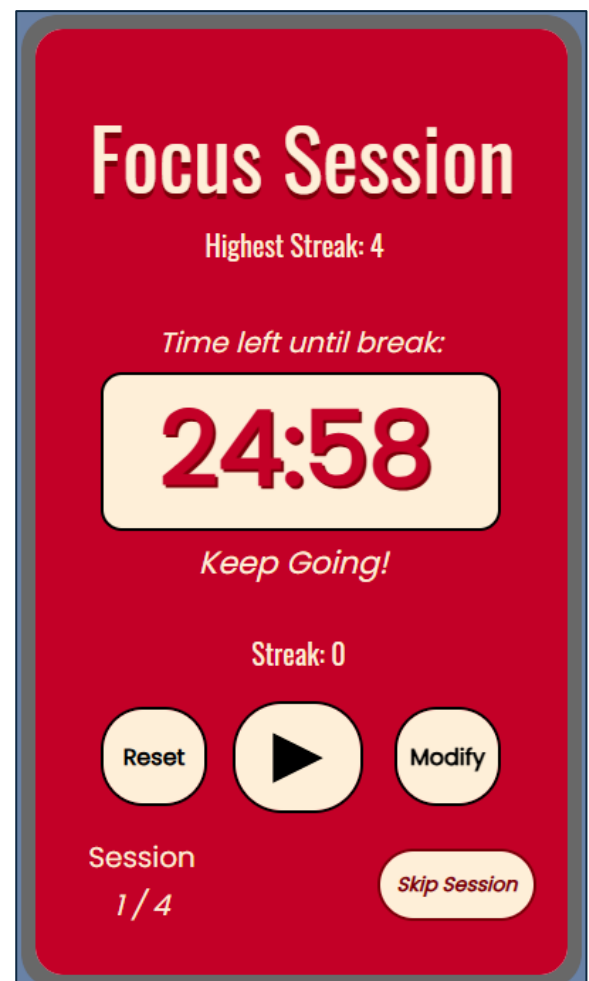
Based on the review of the first version of my app, I have added/adjusted features and elements in my app. See below for details on each adjustment.

Layout Adjustment

I adjusted the layout on all timer screens (all but modify screen) in the way reviewer 1 suggested; Having the Reset, Play/Pause, and Modify buttons towards the bottom of the screen. This allows easier use for users as they will be on mobile devices, as when they are holding their phone, they can easily press the buttons with one hand now.

Furthermore, it also breaks up elements of my app, giving it a more appealing look.

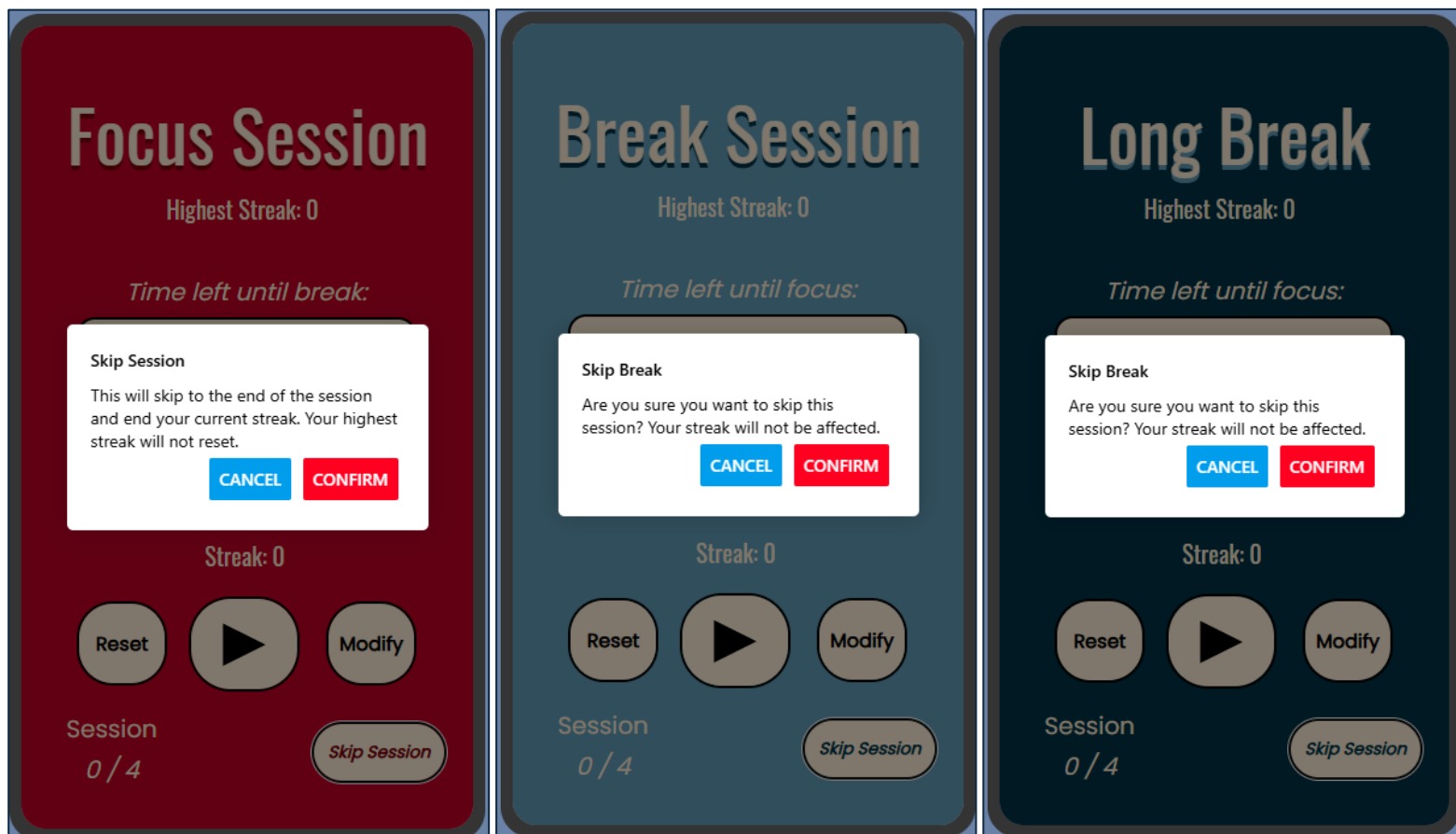
See comparison below between the old and new version (*please note and ignore the white borders around the elements in the old version as this was the editor view of the screen*):



Skip Session Confirmation

I have added a pop-up alert that asks the user to confirm if they want to skip the current session. This was an idea suggested by reviewer 1 as they accidentally clicked the skip button when they did not want to skip the session. This feature prevents this issue for future users. For more information see [here](#).

Screenshots of alerts below for the Focus session and the two Break sessions:

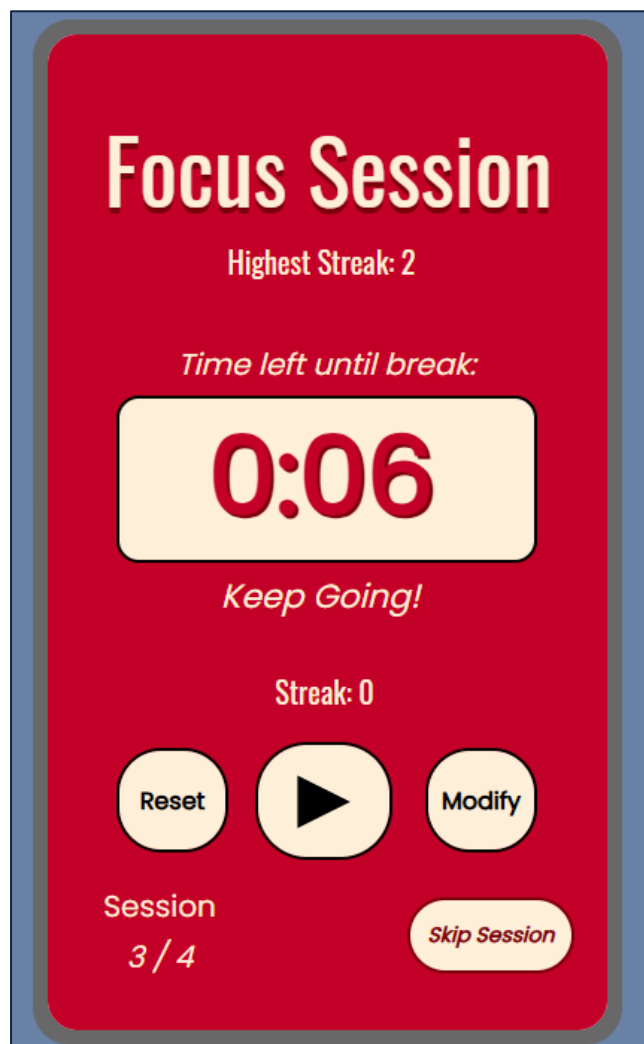


Streak Highscore Feature

As the second reviewer suggested, I have implemented a Streak Highscore counter/feature to my app. This stores the user's highest streak on their device; Therefore, it is saved even after the user shuts the app (the same was their current streak is stored). For more details, see [here](#).

In order to find out how to store variables to the user's mobile device, I had to watch one of Thunkable's tutorials: <https://docs.thunkable.com/blocks/blocks/variables-overview>

This counter displays the user's all time highest streak, as long as they don't delete their app data on their phone or choose to reset it (see [final notes](#) for more detail).



Device Vibrate Feature

As mentioned in the first version of my app's review, to further ensure the user notices the current timer is over, I have added code that causes the mobile device to vibrate. This was a suggestion from the second reviewer I had for my app. For more information, please see [here](#).

I implemented this by using Thunkable's built-in code to make the device vibrate and placed it in the code after the timer finishes (see attached Code Listing document).

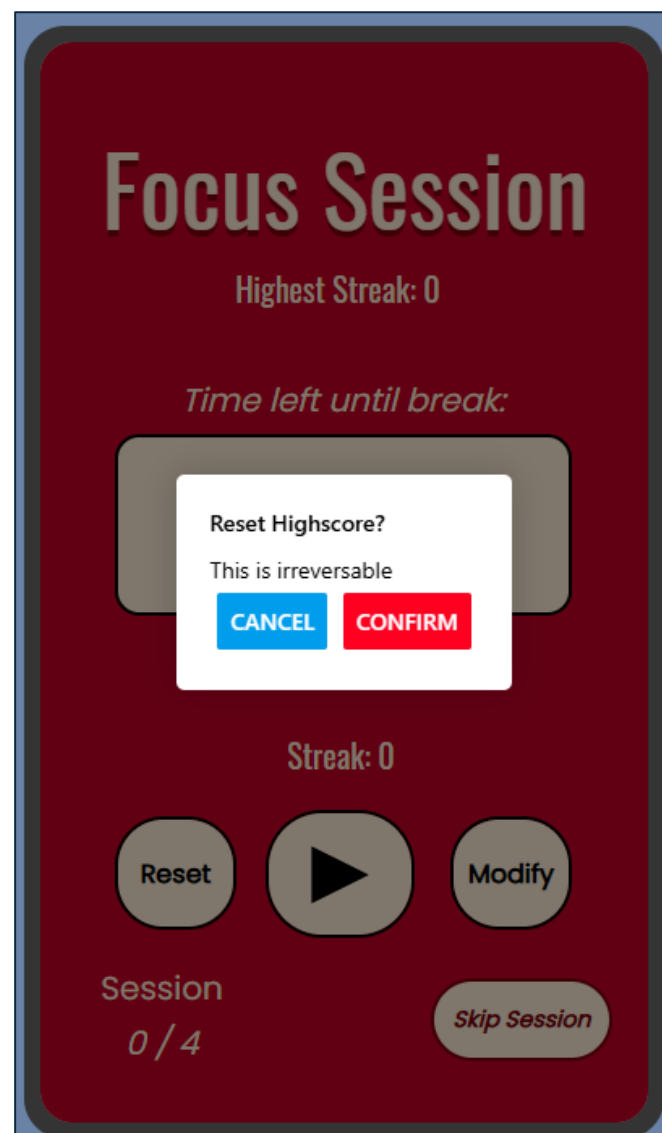
Final Notes

My app still fully works as I originally intended it to. It successfully passed all the functionality test criteria again.

Please see MP4 file: “**No Name Productivity - Version 2.0 Screen Recording - 2026-03-22**” for the overall test and app function (*please note I have adjusted the modify screen’s slider minimum values to allow a simulation of a timer running out quickly without being skipped*).

Reset Highest Streak option:

I also implemented a reset highscore streak option to allow me to test the streak highscore feature easier, including having an alert in case someone was to misclick the reset streak highscore button (see below). This is triggered by the user clicking on the text displaying their highest streak.



Evaluation

Now that I have designed, developed and optimised my app, I will now evaluate it.

Design Evaluation

*Below is the evaluation for the **design** of my app.*

Overview:

I successfully designed my app, planning what requirements (both functional and non-functional, including gamification features), colour scheme, and typography it would have. On top of this, I took several accessibility needs into consideration to ensure my app had an inclusive design. After this I made a wireframe for how my app would loosely function, visually representing the layout of each screen.

Finally, I justified and confirmed how my app's design met all the requirements identified in the scenario.

Afterwards, my app was reviewed by two people who gave feedback on what they liked, disliked, and what improvements I could add to my app based on my design. When I got the feedback from the two people, I had a few issues; However, I was able to resolve them and still get useful feedback towards my app, including what features I should've implemented when I developed my app. For more information on this, please see [here](#).

Successes:

I successfully designed my app, meeting the requirements specified in the assignment brief's scenario.

Issues:

I had some issues when trying to gather feedback on my app's design, however, I was able to resolve it and get very useful feedback.

Improvements:

If I were to do this project again, or a similar project in the future, I would ensure that I have clear questions or specifically state what feedback I would like when gathering feedback. I could do this by making a survey or explicitly specifying to reviewers beforehand.

Development & Optimisation Evaluation

*Below is the evaluation for the **development** of my app.*

Overview:

Development

Before the development of my app, I had to create a test plan that I would later test my app against to ensure that all features and elements of it worked as intended. After I had done this, I went on to start developing my app.

When developing my app, I encountered a lot of bugs and issues (*please see [Creating the App](#)*). These issues mainly stemmed from me creating my code in an inefficient and poorly maintainable way; However, I was able to successfully fix all the issues I had and created a fully functional app based on my designs and the feedback I got.

After this I tested my app against my test plan and was pleased to find it met all the criteria I had planned for.

Finally, I had my developed app reviewed by the same people as before. These people gave me feedback on what I did well, and suggested a few features and changes for the final, optimised version of my app.

Optimisation

After the reviews, I updated my app and modified it to implement the features and changes suggested by the feedback I was given. This was very beneficial for my app as it allowed easier use for users and made my app more appealing in visuals and in the additional features I implemented. *For more detail, please see [My Optimised App Development](#)*.

I then ensured that my app still met all the criteria and functioned the same. Again, I was pleased that it still did.

Successes:

I successfully developed and optimised my app, meeting the requirements specified in the assignment brief's scenario.

Issues:

I had some issues when trying to create my app due to me originally developing it in an unmaintainable way. I then resolved this by changing the way my app was coded.

Improvements:

If I were to do this project again, or a similar project in the future, I would ensure that the app (or other program) is developed in an efficient and maintainable manner from the beginning. This would reduce the chances of bugs and issues occurring later down the line, speeding up development.

Maintenance Evaluation

Overview:

I was able to successfully maintain my app while implementing new features and making changes, however, a lot of issues and bugs occurred throughout this.

As stated earlier, if I were to do this or a similar project again in the future, I would ensure that I developed the project in a maintainable and efficient way, to reduce issues and problems down the line.

In addition, when my app was stress tested, it stopped functioning correctly and softlocked the user on a break screen unless they were to restart the app (see [here](#)). This is an issue I would have to fix to maintain my app's functionality, otherwise it would likely put users off my app.

Finally, to make my app more maintainable, I should update my code to ensure all screens are as efficient as possible. Currently, the break session screens are not as efficient as the focus session screen; This causes issues when a new feature is added to the app as it often causes bugs. Furthermore, to add a feature, I must add it to all three session screens; This takes a lot of time to update, making it harder to maintain. This is something I would have to keep in mind if I were to further develop my app in the future.

Final App Evaluation

Below is the evaluation of my mobile app as a whole.

App Effectiveness

At meeting the assignment brief's criteria:

My app has been effective at meeting the assignment brief's criteria:

- It helps support students with their *general* study skills – This is because my app is a style of pomodoro timer, which are popular for improving study skills for students. *For more detail see [here](#).*
- It utilises gamification to help motivate them to engage with it – My app has features such as streaks and highscores to encourage the students to engage with it, it also has rewarding elements by giving them a short break after every focus session, and long breaks when they finish their set interval of focus sessions. Furthermore, my app allows for customisation of the duration of the timers, and the interval at which they get rewarded by a long break; This not only gives them the freedom & flexibility at how they wish to study, but will also improve their independence as they manage themselves and their studies through using the app. *For more detail see [here](#).*
- My app is also visually appealing – I put time into choosing the correct colour scheme, typography, and layout for my app to make it visually appealing. I then adjusted the layout and design further based on based on feedback received to ensure that my app would be visually appealing to its users. *For more detail see [here](#).*
- My app is intuitive to choose and does not require additional guidance – This is because the elements are self-explanatory and even labelled to tell the user what they do. Other elements, which have knock-on effects that might not be clear, have pop-up alerts that clarify important information to the user (such as the skip session button). This ensures that the user won't need additional guidance on how the app works. However, if users do not know how a pomodoro timer works, they might be confused at first on how my app works; To fix this, I could create a start screen that briefly explains what my app does, and how it works; This would be a future development to improve my app further. *For more detail see [here](#).*
- My app runs on an Android or iOS mobile device – Because I built my app using Thunkable, I can release my app for Android and iOS devices as that is one of their main features. *For more detail see [here](#).*

In comparison to alternative solutions:

There are many similar, existing version of my app already; However, I have ensured that my app only includes necessary features that will encourage users (primarily students) to improve their study skills and independence, including time management. Because of this, I have also ensured that the app is not overwhelming to the user from not having unnecessary elements.

Another unique feature of my app is that the colour of the background changes based on what session the user is in (focus, break, or long break); This has a positive psychological effect on them as explained [here](#).

Future Developments:

If I were to further develop my app in the future, I would add some more features. I have gone into detail about these below:

- **Auto Start Timer** – This option would allow the user to toggle on or off each session and its timer starting automatically after the prior session has ended; As opposed to waiting for the user to press next, and the session timer starting paused (as it currently does). This would give the user more control over the app, and if they were to use this option, they would be less likely to get distracted or stop studying between sessions.
- **User Praise update** – As an additional feature to motivate the user more, the praise/motivational messages that appear on-screen could be randomised from a set list, rather than just being the same message each time. This makes the message feel more meaningful and unique to the user, encouraging them further to continue their studies/work.
- **Timer Font Rewards** – As another additional feature, timer font rewards would encourage the user to keep their streak (which already encourages them to use the app regularly) by updating the timers' font every time they reach a streak milestone, such as when their streak gets to 10, 20, 40, etc. This would further amplify the gamification aspect of my app and again, will encourage app use. For more detail, see [here](#).

Project Plan / Development Log

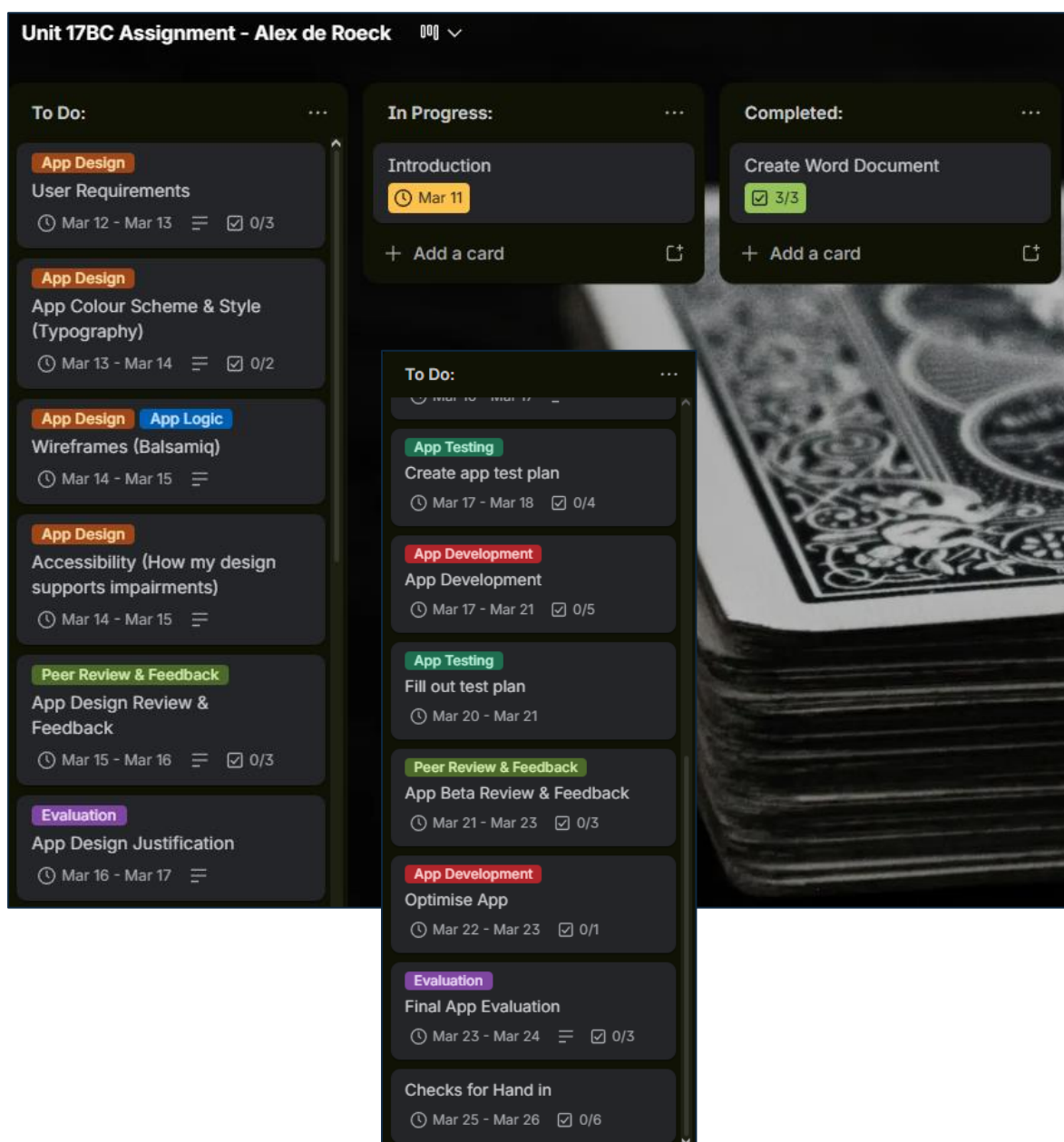
Throughout this coursework I will follow a plan to ensure that I complete the project in time for the hand in date. *To do this, I have created a plan using [Trello](#), creating tasks that I will move between “To Do”, “In Progress”, and “Completed” lists.*

11/03/2026:

I have created the plan & tasks for my coursework.

I have also set due dates for them, and sub tasks to make sure I don't miss parts of my write-up.

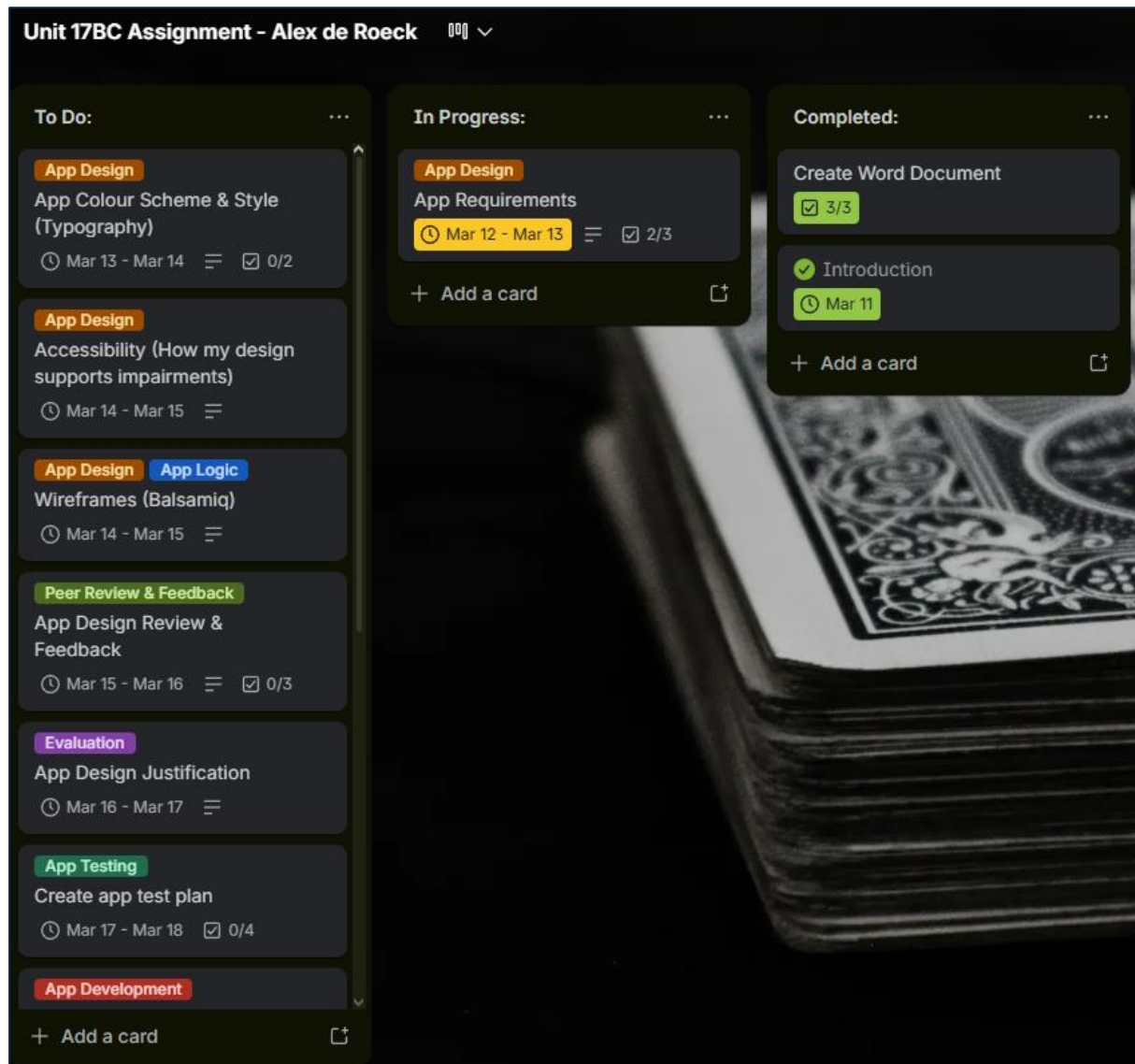
I have started to write the introduction of the write-up.



12/03/2026:

Finished the introduction for the write-up.

Almost finished the app requirements.

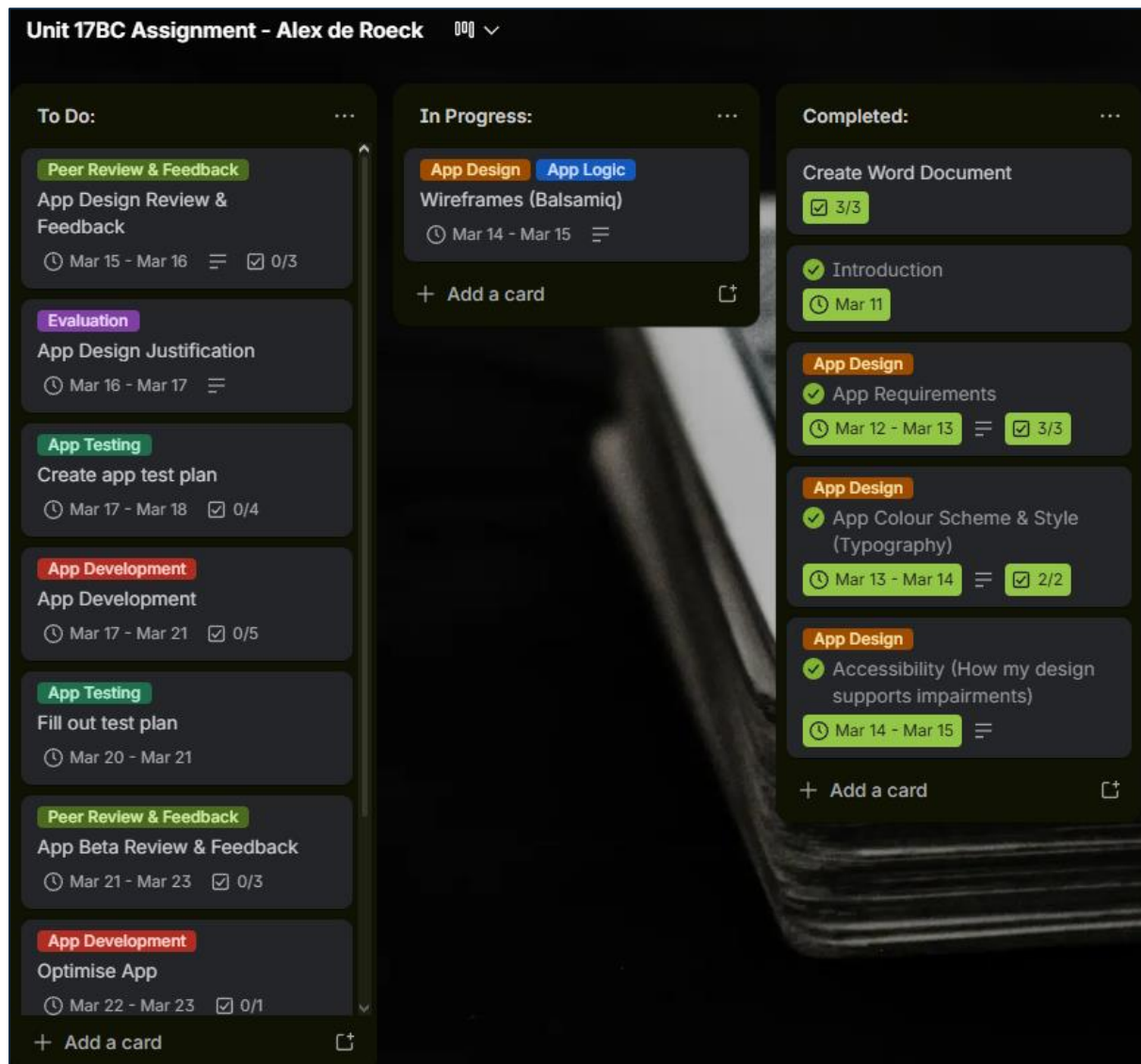


13/03/2026:

Finished App Requirements, App Colour Scheme & Style (Typography), and Accessibility.

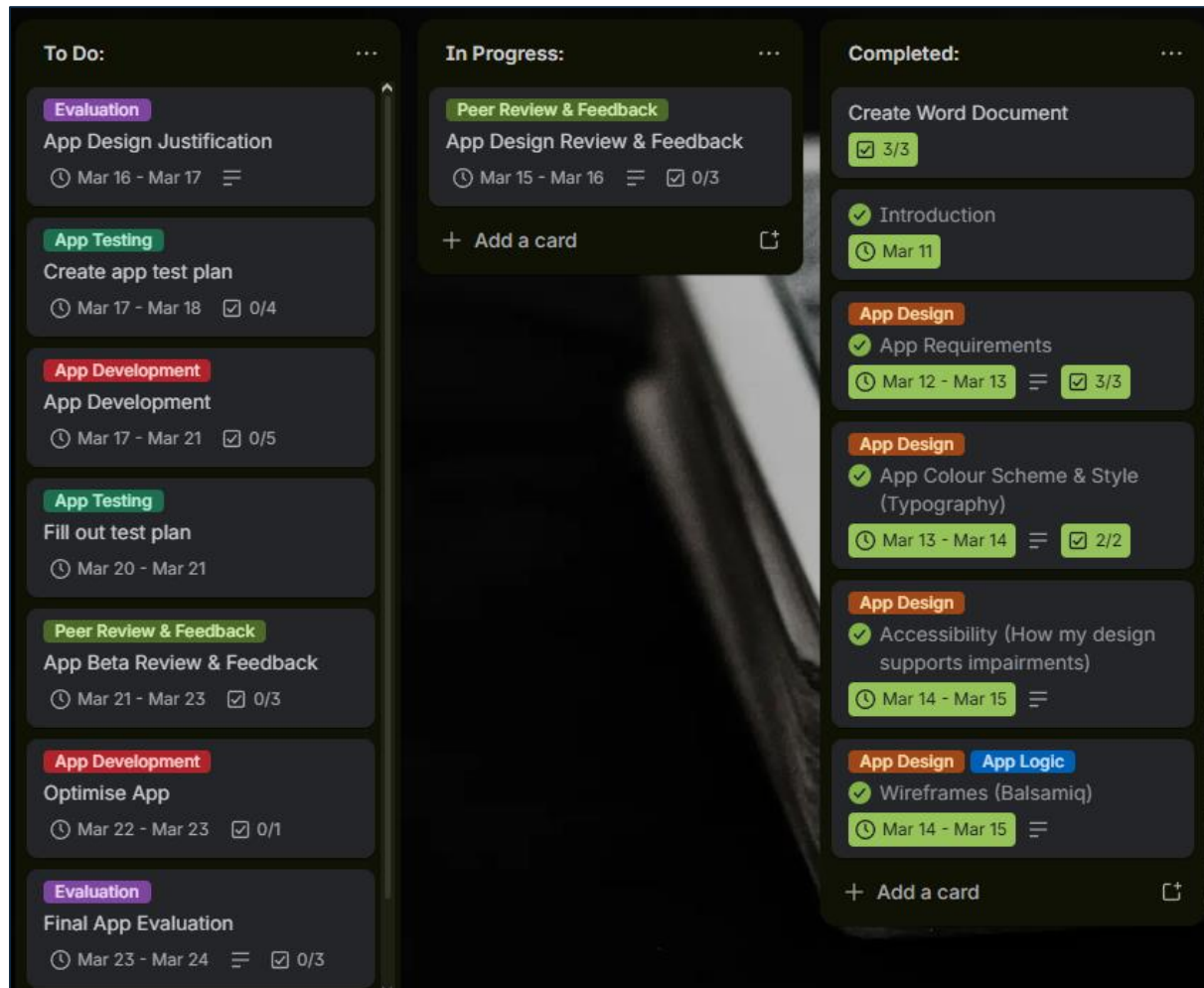
I am going to do the wireframes tomorrow and hopefully make a start or potentially finish App Design Review & Feedback.

I am currently ahead of schedule.



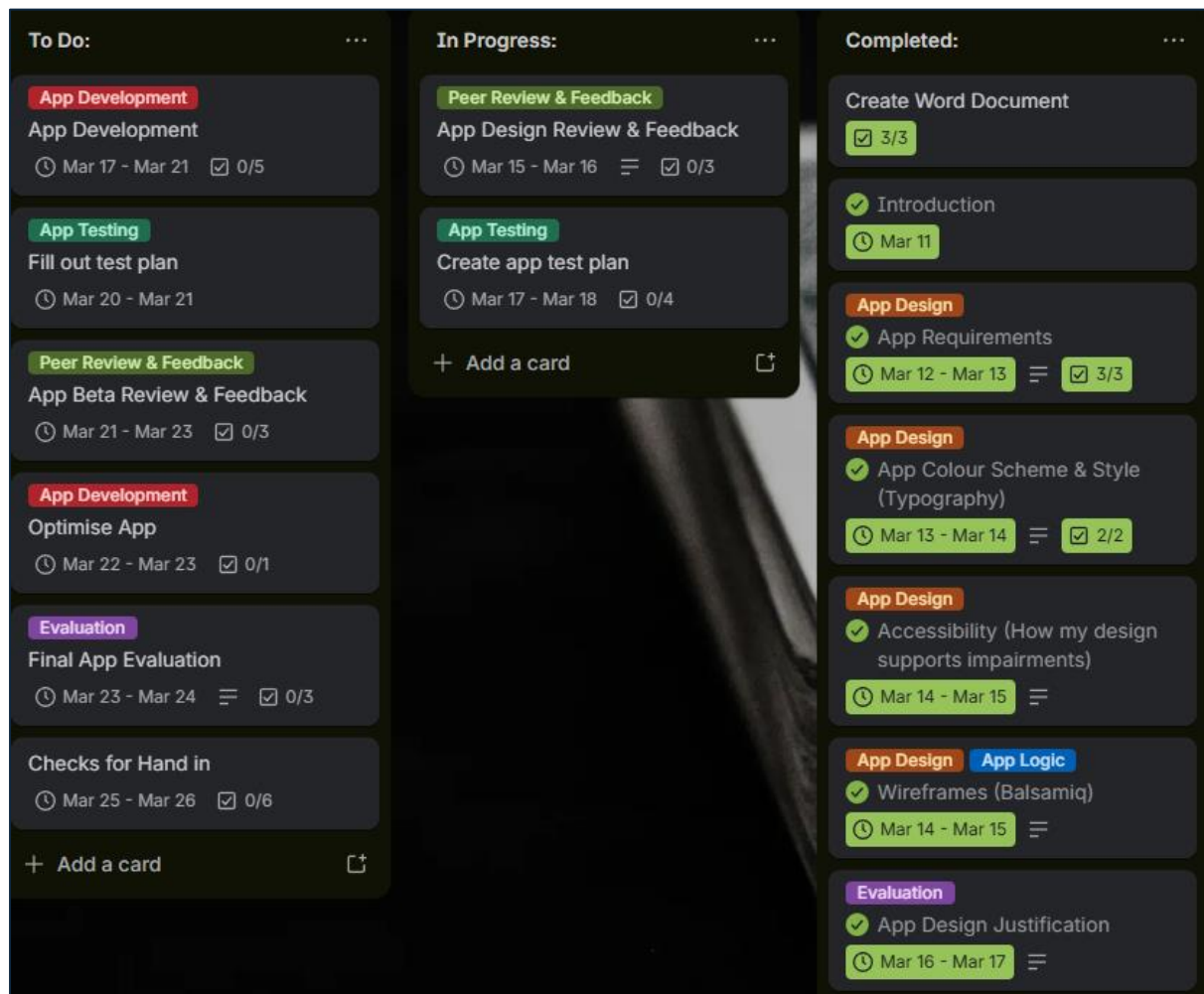
14/03/2026:

Completed the wireframes for my app. This took longer than expected, however, I am still on time with my schedule for my coursework. I was unable to start the App Design Review & Feedback.



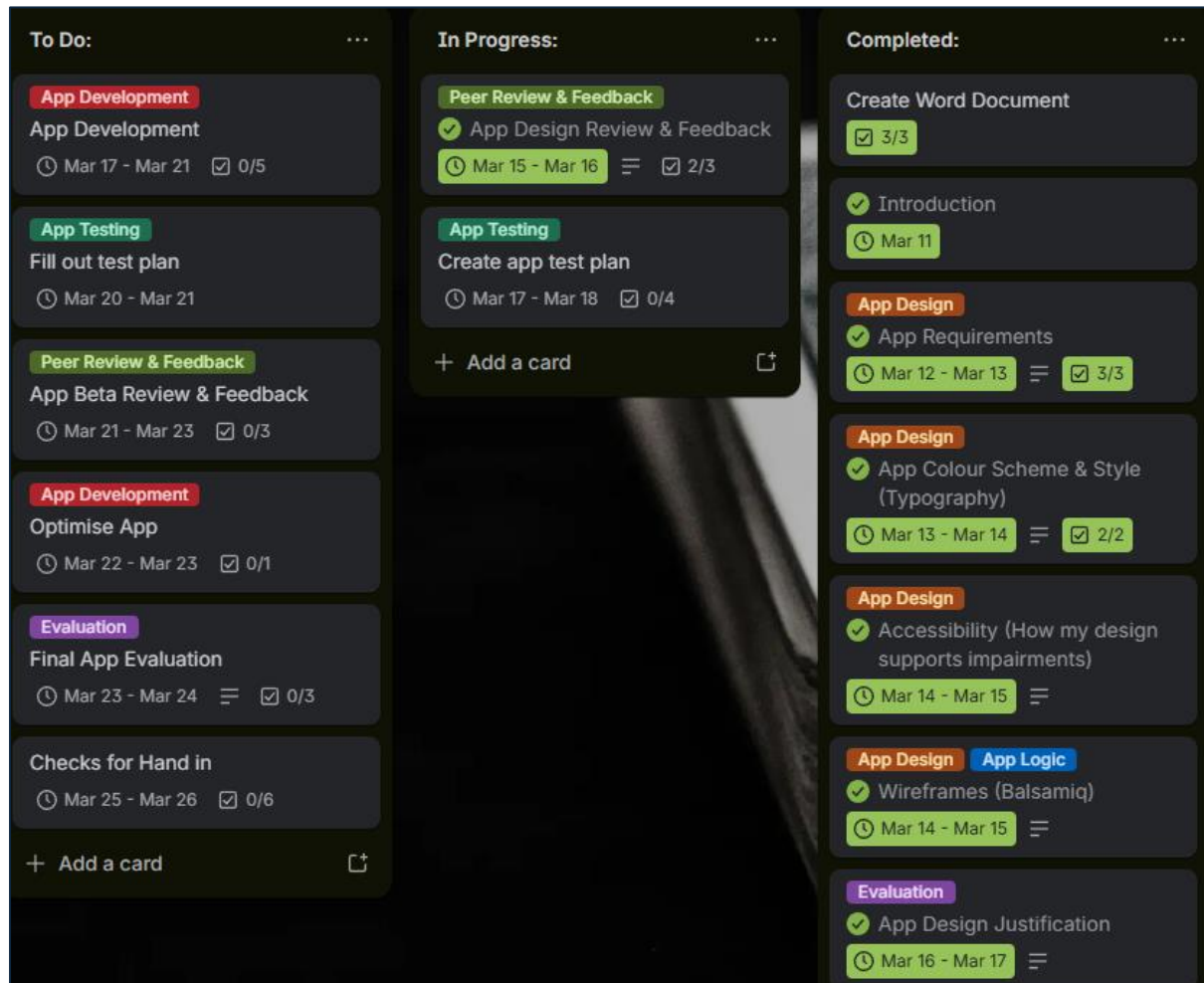
15/03/2026:

Completed App Design Justification. Potentially receiving feedback this evening.



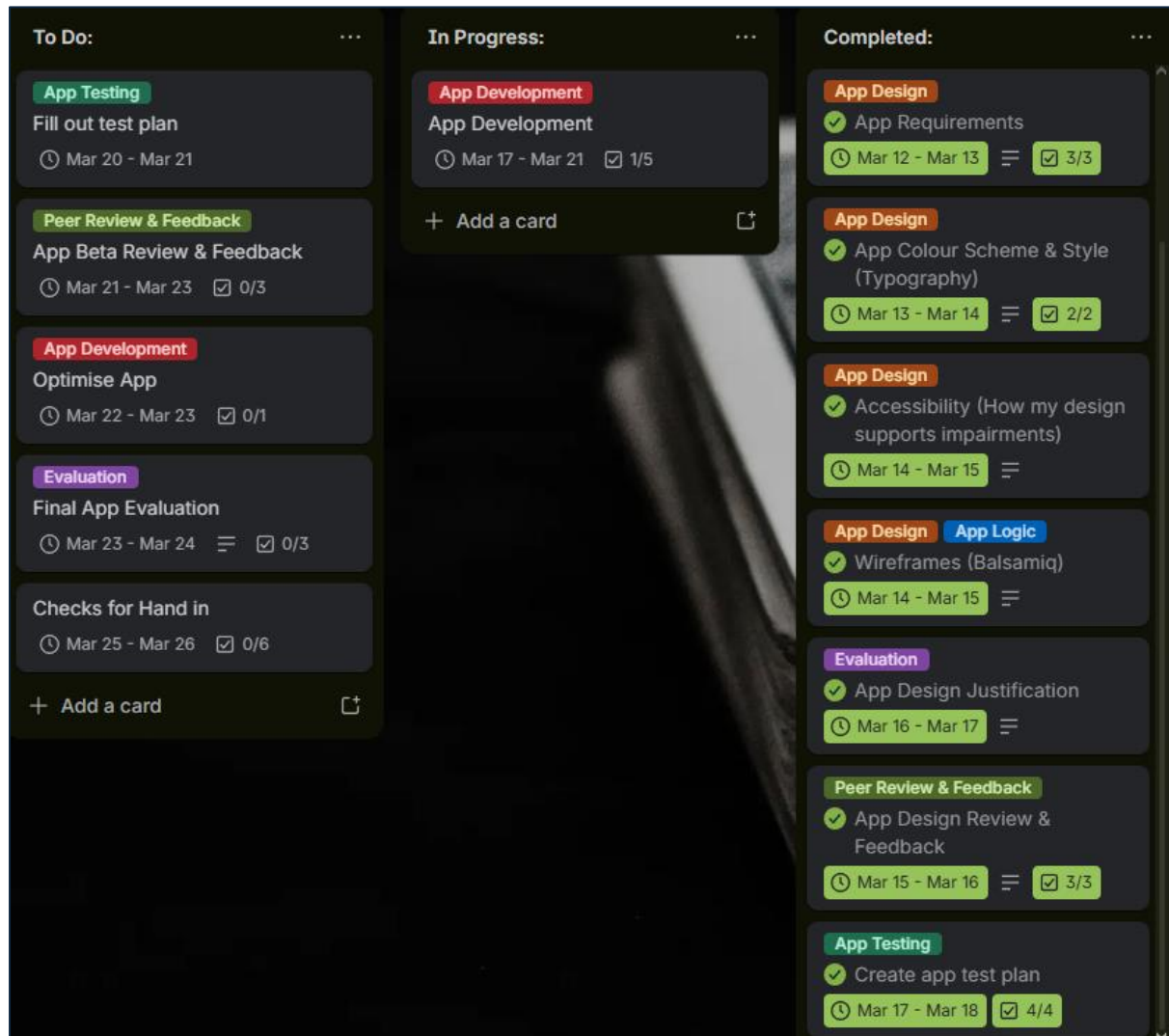
16/03/2026:

Got and reviewed feedback. Still need to double check its finished tomorrow. I've been set back on my coursework as I did not consider time that I would need to spend revising for my other subjects' mock exams. However, I am still on track for finishing on time.



17/03/2026:

- Finished the write-up for the App Design Review & Feedback
- Created my app test plan
- Wrote up for reasoning behind app development environment choice
- Started to develop my app

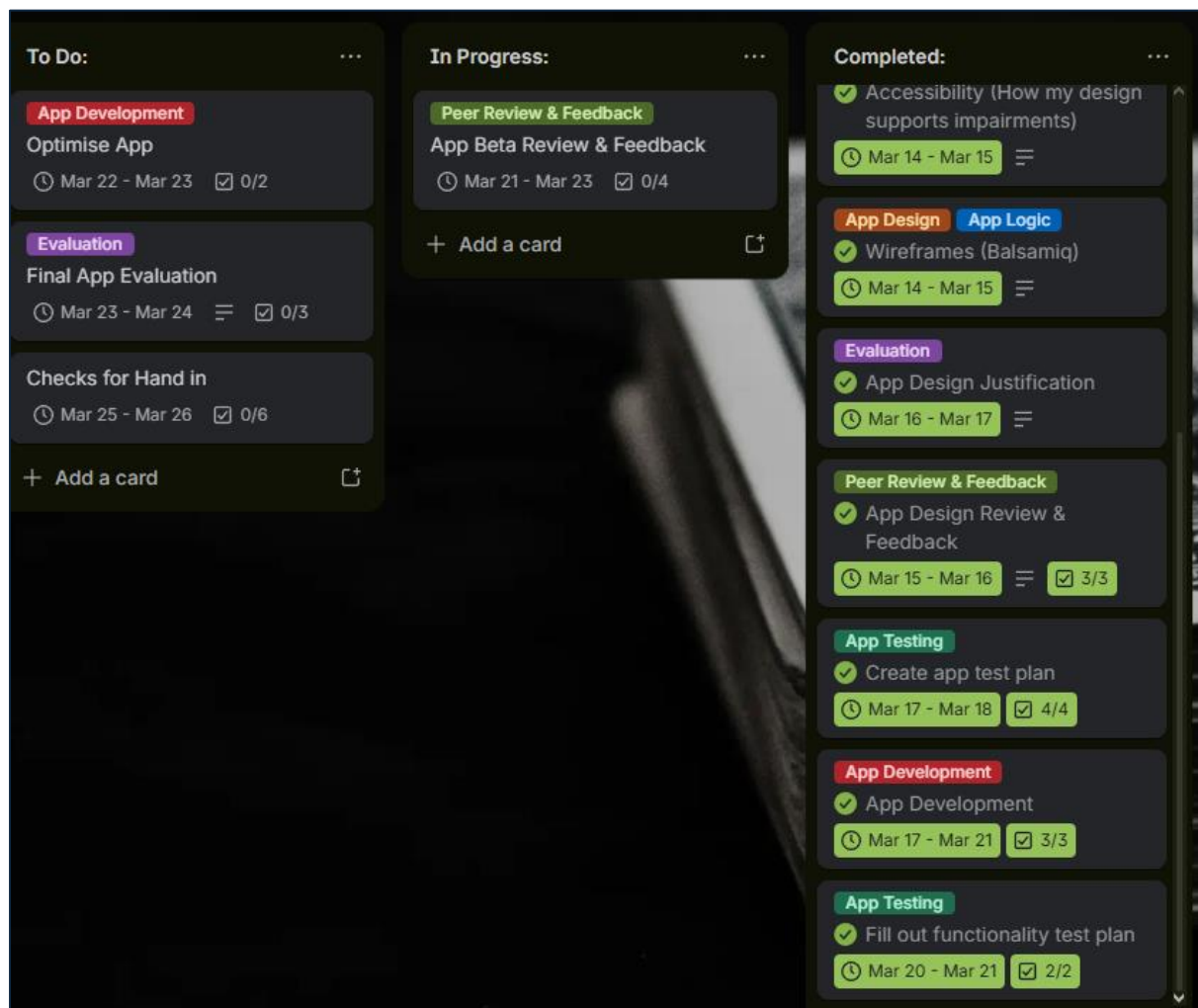


18–21/03/2026:

Developed my mobile app.

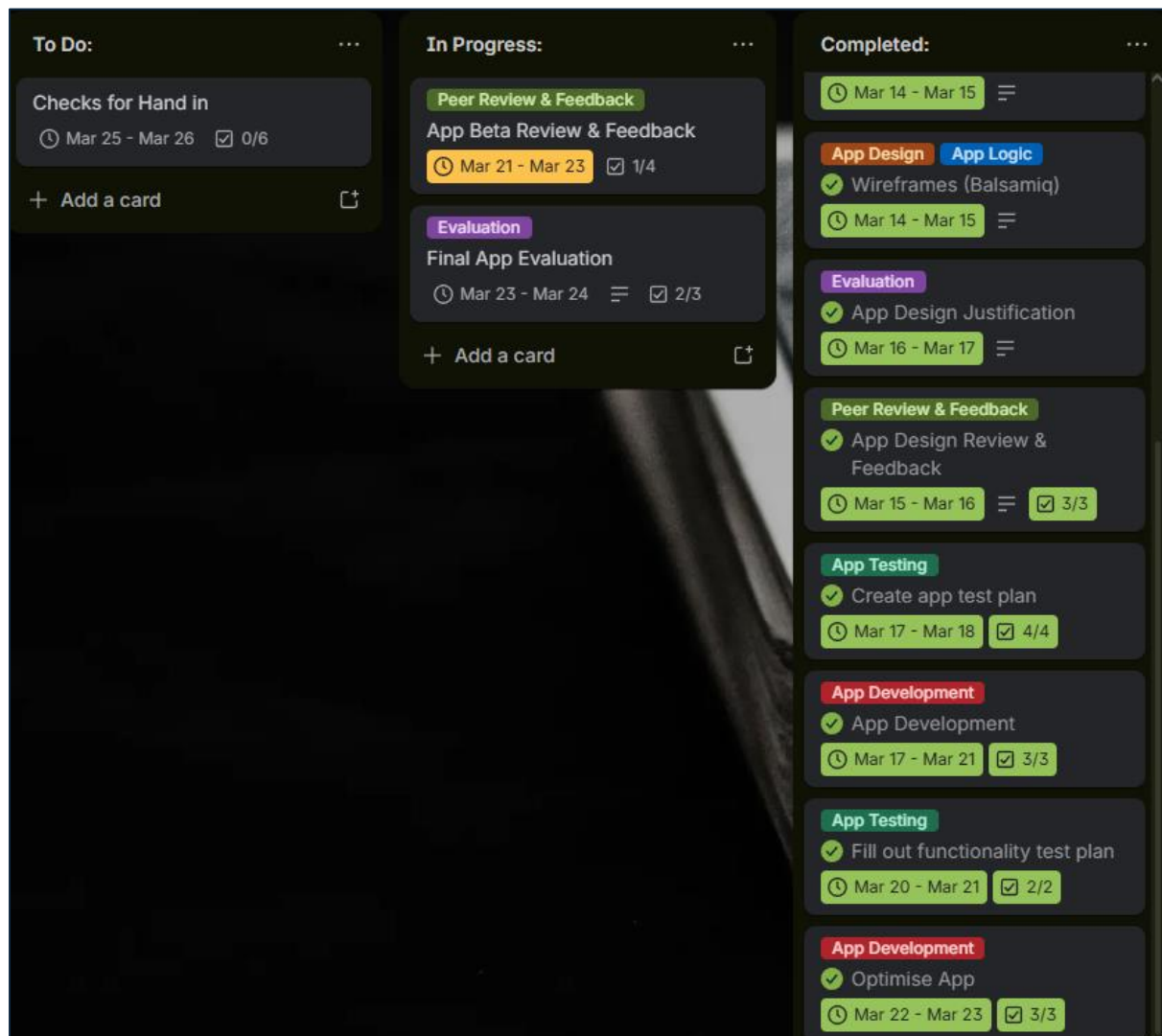
- Finished Developing my mobile app
- Filled out the functionality test plan
- Created code listing document (for version 1 of my app)
- Screen recorded above testing of full functionality of my app

See [Creating the App](#) for more details on all the above.



22/03/2026:

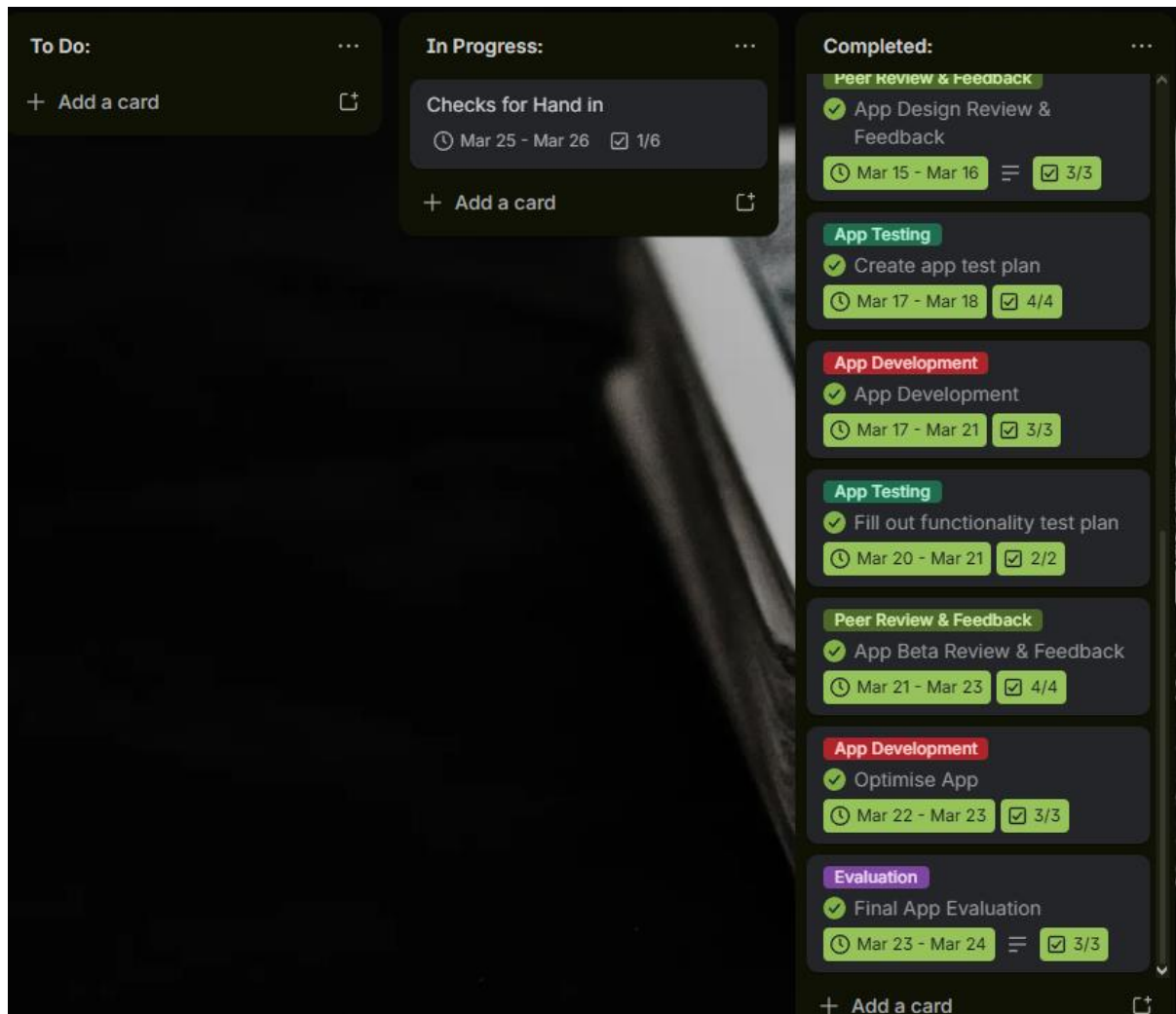
Today I got some feedback, optimised my app, and did most of the evaluation.



23/03/2026:

Finished the project evaluation. Double checked SPaG of this write-up document.

Tomorrow, I will do any final double checks and additional file formatting needed before handing in my assignment.



24/03/2026:

Final Hand in.

I double checked the assignment, will create a PDF version of my write-up and code listing, and zip all video files.

References:

[“The Psychology of Color in UI/UX Design” Article](#) – Used [here](#)

[“Fiery Ocean” Coolers](#) colour palette – Used [here](#)

[“Balsamiq Wireframes” App](#) – Used [here](#)

[“Trello” Planner website](#) – Used [here](#)

[“Thunkable” App building website](#) – Used [here](#)